Notes on Master Studies

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Abstract

"En remontant chez moi pour y passer la soirée à travailler de mon mieux, je me disais que le monde n'est pas construit pour l'équilibre. Le monde est désordre. L'équilibre n'est pas la règle, c'est l'exception."
G.Duhamel, Maitres, 1937

Chapter 1

Random Matrix Theory

Random Matrices can be though of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such a random matrix. With the help of some mathematical tools such as orthogonal polynomials one can expresses the equilibrium measures for the eigenvalues asymptotically.

1.1 Random Matrices

The chapter that follows is heavily based (to be very generous with my writing) on the incredibly organized and written Thesis [2]. To the author, my best wishes. The idea of this chapter is to motivate and introduce the concepts we will touch on later for the main part of the work. Also, if it is of any worth, this chapter serves as a good recap for any reader and a disclaimer of all the notation and conventions we will use in this work. What follows is my best attempt in introducing the part of Random Matrices that are pertinent to my scope of work and are of general interested, albeit inside my limitations.

1.1.1 Random Matrices Ensembles

Borrowing some concepts from physics, an ensemble can be though as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. There is a analogy to be made with ensembles of thermodynamic: When considering sets of matrices with equal probability - in some sense to be more clear soon - we get the Gibbs weight for the probability distribution as we would in the *Canonical Ensemble* of constant temperature. More concretely, in the context of Random Matrix Theory

Definition 1.1.1. *A random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is a set of matrices of fixed size equipped with a probability density function (p.d.f) $\mathcal{P} : \mathcal{S} \to [0, \infty)$. If we say that a matrix $\hat{X}$ is drawn from the random ensemble $\mathcal{E}$, we are saying that it is a random variable assuming values in $\mathcal{S}$ with p.d.f. $\mathcal{P}$.*

A commonplace thing to do when defining ensembles is to prescribe the p.d.f's $\mathcal{P}_{i,j}$ on the independent entries $X_{i,j}$ of a representative matrix $\hat{X} \in \mathcal{E}$. In such cases, we say that

$$\mathcal{P}(\hat{X}) = \prod_{i,j | X_{i,j} independent} \mathcal{P}_{i,j}(X_{i,j}). \tag{1.1.1}$$

With this definition we can already define many ensembles of interest. For many, the set of matrices is described by applying constraints to the set of real matrices $\mathbb{M}_{m,n}(\mathbb{R})$, the set of complex matrices $\mathbb{M}_{m,n}(\mathbb{C})$ or the set of quaternionic matrices $\mathbb{M}_{m,n}(\text{He})$. Then, oftentimes, $\mathcal{S}$ is simply the set of symmetric, hermitian or skew-symmetric matrices. When it comes to determining the p.d.f $\mathcal{P}$ on $\mathcal{S}$, there are many options that have been worked on. A simple example is that of *Bernoulli matrices* $\hat{B}$ whose entries $B_{i,j}$ are all independently and identically distributed Bernoulli random variables ± 1: $\mathcal{S} = \mathbb{M}_{n,n}(\mathbb{R})$ and

$$P_{i,j}(B_{i,j}) = \frac{1}{2} \left[\delta(B_{i,j} - 1) + \delta(B_{i,j} + 1) \right].$$

If we instead require $\hat{B}$ to be symmetric, its diagonal entries to be zero, and its independent entries have p.d.f.

$$P_{i,j}(B_{i,j}) = \frac{c}{n}\delta(B_{i,j}-1) + \left(1 - \frac{c}{n}\delta\left(B_{i,j}\right)\right),$$

then $\hat{B}$ belongs to the ensemble of *Erdos-Renyi.* Consider now one of the most important ensembles we will define.

Definition 1.1.2. *(Ginibre Ensemble). The $m \times n$ real, complex, and quaternionic Ginibre Ensembles are respectively $\mathbb{M}_{m,n}(\mathbb{R}), \mathbb{M}_{m,n}(\mathbb{C}), \mathbb{M}_{m,n}(\mathrm{He})$ with p.d.f*

$$\mathcal{P}^{(Gin)}(\hat{G}) = \pi^{-mn\frac{\beta}{2}} \exp\left(-\operatorname{Tr}(\hat{G}^\dagger \hat{G})\right) = \prod_{i=1}^{m}\prod_{j=1}^{n} \pi^{\frac{\beta}{2}} \exp\left(-|G_{i,j}|^2\right), \tag{1.1.2}$$

defined in respect to the Lebesge measure

$$\mathrm{d}\hat{G} = \prod_{i=1}^{m}\prod_{j=1}^{n}\prod_{s=1}^{\beta} \mathrm{d}G_{i,j}^{(s)}. \tag{1.1.3}$$

The coefficient β is called the Dyson index and counts the generic number of real components of each matrix entry $G_{i,j}$, so that $\beta = 1, 2, 4$ refers to real, complex and quaternionic entries. These are the Ginibre orthogonal, unitary and symplectic ensembles.

For most applications there are three ideologies that stand out. The first one is to assume little on the system and focus on universal results. This results in the study of the *Wigner matrix ensembles*: Let $\hat{X}$ be drawn from either the real symmetric, complex hermitian or quaternionic skew-symmetric $n \times n$ matrix ensemble. Then its entries $X_{i,j}$ are independently distributed with mean zero and bounded moments. Also, its diagonal entries are equally distributed as are its upper triangular entries. Many results hold for independently and identically distributed centered random variables. To construct the general form of such matrices take the matrix $\hat{Y} = \{Y_n\}_{i,j}$ with $Y_{i,j} = Y_{j,i}^\dagger$ and such that

1. $\{Y_{i,j}\}_{1 \le i \le j \le n}$ are independent;
2. $\{Y_{i,j}\}$ are identically distributed. $\{Y_{i,i}\}$ are identically distributed;
3. $\mathbb{E}(Y_{i,j}) = 0$ and $\mathbb{E}(Y_{i,j}^k) < \infty$, i.e, $r_k = \max\left(\mathbb{E}(Y_{i,j}^k), \mathbb{E}(Y_{i,j}^k)\right) < \infty$.

Definition 1.1.3. *Let every $\{Y_n\}_{i,j}$ be as before and set $r_2 < \infty$ with $\mathbb{E}(Y_{1,2}^2) > 0$. Then, we say $\hat{Y}$ is a Wigner Matrix. Furthermore, if $\hat{X}_n = n^{-\frac{1}{2}}\hat{Y}_n$, $\hat{X}$ is what we call a normalized Wigner Matrix.*

The scale and moment assumption taken are not as unnatural as they may seem at first. To properly understand this definition, note that if $\lambda_1^2, \lambda_2^2, \ldots, \lambda_n^2$ are the ordered eigenvalues of $\{Y_n\}_{i,j}$, we can estimate that

$$\lambda_1^2 + \cdots + \lambda_n^2 \le n \max_{j=1}^{n}(\lambda_j^2) = n \max(\lambda_1^2, \lambda_n^2) \tag{1.1.4}$$

therefore, if we want the eigenvalues to be limited when we take large matrices we should impose that $1/n \cdot (\lambda_1^2 + \cdots + \lambda_n^2)$ must be limited. This is the Hilbert-Schmidt or Frobenius norm for these normal matrices:

$$||\hat{Y}||_{HS} = \frac{1}{n}\sum_{i=1}^{n}\lambda_i^2 = \frac{1}{n}\sum_{i,j} Y_{i,j}^2 = \frac{1}{n}Tr(\hat{Y}^2) \tag{1.1.5}$$

then if we only take $\hat{X}_n = \alpha_n \hat{Y}_n$

$$\begin{aligned}\mathbb{E}(||\hat{X}_n||_{HS}) &= \alpha_n^2 \mathbb{E}(||\hat{Y}_n||_{HS}) \\ &= \frac{\alpha_n^2}{n}\sum_{i,j}\mathbb{E}(Y_{i,j}^2) \\ &= \alpha_n^2\mathbb{E}(Y_{1,1}^2) + (n-1)\alpha_n^2\mathbb{E}(Y_{1,2}^2)\end{aligned} \tag{1.1.6}$$

then it is clear that there are two cases

1. $\mathbb{E}(Y_{1,2}^2) = 0$: This is the non-interesting case where $\alpha_n \approx 1$

2. $\mathbb{E}(Y_{1,2}^2) > 0$: Here, we must take $\alpha_n \approx n^{-\frac{1}{2}}$

Now, it is likely more clear the motivation of the definition: limit the eigenvalues to a compact region. As the norm is limited, also are the eigenvalues of the scaled matrix.

The second ideology is assuming equal probability on each state os the system - this is a common assumption in thermodynamics. So this approach is to assume that every matrix is equally likely.

Observation 1.1.4. *We have focused on random matrix ensembles distributed according to p.d.f.s $\mathcal{P}$ defined with respect to the flat Lebesgue measure induced by the canonical embedding of $\mathcal{S}$ into Euclidean space. Note here, which was not obvious for me, that the measures ν we define are given in terms of the p.d.f.s $\mathcal{P}$ and the Lebesgue measure μ by Radom-Nikodin*

$$\nu(A) = \int_A \mathcal{P}\mathrm{d}\mu, \tag{1.1.7}$$

or more simply, $\mathrm{d}\nu = \mathcal{P}\mathrm{d}\mu$.

Theorem 1.1.5. *(Haar, compact case). Given a compact group $\mathcal{S}$, let $U \in \mathcal{S}$ be a generic variable. There exists a unique (up to normalization measure) $\mathrm{d}\mu(U)$ on $\mathcal{S}$, called the* ***Haar measure*** *such that*

$$\int_{\mathcal{S}} \mathrm{d}\mu(U) < \infty \tag{1.1.8}$$

and for any fixed $U_0 \in \mathcal{S}$,

$$\mathrm{d}\mu(U_0U) = \mathrm{d}\mu(U) = \mathrm{d}\mu(UU_0),$$

that is, finite for compact sets and invariant by left and right actions. If $\mathrm{d}\mu(U)$ integrates to unity in $\mathcal{S}$ it is referred as the ***Haar probability measure****.*

Definition 1.1.6. *The $n \times n$ circular unitary ensemble (CUE) is the unitary group $U(n)$ with its Haar measure. If $U \in$ CUE(n), then the symmetric unitary matrix U^TU represents the $n \times n$ circular orthogonal ensemble (COE); we have $\mathcal{S} \cong U(n)/O(n)$. Note that if $U \in U(n)$ and $O \in O(n)$, we have that $U^TU = (OU)^T(OU)$, that is, U and OU are in the same co-class.*

The third ideology is that of working with algebraic combinations of matrices whose entries are independently and identically distributed. This will not be as much discussed.

1.1.2 Some quantities

To study random matrices, we need to determine which quantities related to the matrix we want to measure. This may be the expectation of some function on the matrices or even on the eigenvalues and eigenvector, if they exist. Assume a matrix is such that is able to be determined by eigenvalues and eigenvectors. While eigenvector may be of interest in some cases, eigenvalue statistics have been considerably more explored. Hence, in addition to the p.d.f.s $\mathcal{P}(X)$ of say $n \times n$ random matrices $\hat{X}$, we are also interested in the eigenvalue j.p.d.f. $\mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ of the eigenvalues $\{\lambda_i\}_{i=1}^n$ of $\hat{X}$. The eigenvalue j.p.d.f. is defined to be such that

$$\int_{[a_1,b_1]\times\cdots\times[a_n,b_n]} p(\lambda_1, \lambda_2, \ldots, \lambda_n)\mathrm{d}\lambda_1 \cdots \mathrm{d}\lambda_n \tag{1.1.9}$$

is the probability that $a_i < \lambda_i < b_i$ for every i.

Example 1.1.7. *The eigenvalue j.p.d.f of the $n \times n$ circular ensembles is*

$$\mathtt{p}^{(C)}(\mathrm{e}^{\mathrm{i}\theta_1}, \cdots, \mathrm{e}^{\mathrm{i}\theta_n}; \beta) = \frac{1}{\mathcal{Z}_{n,\beta}^{(C)}} |\Delta_n(\mathrm{e}^{\mathrm{i}\theta})|^\beta \tag{1.1.10}$$

where $Z_{n,\beta}^{(C)}$ is a normalization constant and

$$\Delta_n(\lambda) := \det\left[\lambda_i^{j-1}\right]_{i,j=1}^n = \prod_{1\le i<j\le n} (\lambda_i - \lambda_j) \tag{1.1.11}$$

is the Vandermonde determinant, and $\beta = 1, 2, 4$ correspond to the COE, CUE and CSE, respectively.

Related do the eigenvalue j.p.d.f is the univariate eigenvalue density defined by

$$\mathfrak{p}(\lambda_1; n) := n \int_{\mathbb{R}^{n-1}} \mathfrak{p}(\lambda_1, \lambda_2, \ldots, \lambda_n) \mathrm{d}\lambda_2 \cdots \mathrm{d}\lambda_n, \tag{1.1.12}$$

which is normalized such that the expect number of eigenvalues lying between a and b is given by $\int_a^b \mathfrak{p}(\lambda)\mathrm{d}\lambda$. Note that this is not a p.d.f if $n \neq 1$. Consider now the linear statistic of the eigenvalue density given by

$$\mathfrak{p}(\lambda) = \sum_{i=1}^{n} \langle \delta(\lambda - \lambda_i) \rangle \tag{1.1.13}$$

where the brackets are understood as acting on an operator as

$$\langle \mathcal{O} \rangle = \int_{\mathbb{R}^n} \{ \mathcal{O} \mathfrak{p}(\lambda_1, \cdots, \lambda_n) \} \mathrm{d}\lambda_1 \cdots \lambda_n. \tag{1.1.14}$$

Many times, eigenvalues density are fully characterized by their *spectral moments*

$$m_k := \int_{\mathbb{R}} \lambda^k \mathfrak{p}(\lambda) \mathrm{d}\lambda, \ k \in \mathbb{N}. \tag{1.1.15}$$

Indeed, if we are interested in a linear statistic $\sum_{i=1}^{n} f(\lambda_i)$ of the eigenvalues such that $f(\lambda)$ is analytic at zero with Taylor expansion $f(\lambda) = \sum_{k=0}^{\infty} f_k \lambda^k$, we can compute its expectation to be

$$\left\langle \sum_{i=1}^{n} f(\lambda_i) \right\rangle = \int_{\mathbb{R}} f(\lambda) \mathfrak{p}(\lambda) \mathrm{d}\lambda = \sum_{k=0}^{\infty} f_k m_k \tag{1.1.16}$$

The spectral moments can additionally be expressed as an average with respect ot the eigenvalue j.p.d.f. $\mathfrak{p}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ and the p.d.f $\mathcal{P}(X)$ of the random matrix $\hat{X}$ according to

$$m_k = \left\langle \sum_{i=1}^{n} \lambda_i^k \right\rangle = \left\langle \mathrm{Tr}(\hat{X}^k) \right\rangle_{\mathcal{P}(\hat{X})} \tag{1.1.17}$$

where we define

$$\langle \mathcal{O} \rangle_{\mathcal{P}(\hat{X})} := \int_{\mathcal{S}} \{ \mathcal{O} \mathcal{P}(\hat{X}) \} \mathrm{d}\hat{X} \tag{1.1.18}$$

Moment Generating Function

We can also define an useful tool for studying the spectral moments: The *Moment Generating Function* given by

$$W_1(x) := \sum_{k=0}^{\infty} \frac{m_k}{x^{k+1}}, \tag{1.1.19}$$

which is also know as the *Resolvent* due to it being equal to the *Stieltjes Transform* of the eigenvalue density

$$W_1(x) = \int_{\mathrm{supp}(\mathfrak{p})} \frac{\mathfrak{p}(x)}{x - \lambda} \mathrm{d}\lambda, \quad \text{for } x \in \mathbb{C} \backslash \mathrm{supp}(\mathfrak{p}). \tag{1.1.20}$$

Obtaining a closed form expression for the resolvent allows one to use the *Sokhotski-Plemeji* inversion formula to recover the eigenvalue density

$$\mathfrak{p}(\lambda) = \lim_{\epsilon \to 0} \frac{W_1(\lambda - \mathrm{i}\epsilon) - W_1(\lambda + \mathrm{i}\epsilon)}{2\pi \mathrm{i}} \tag{1.1.21}$$

as long as $\mathfrak{p}(\lambda)$ is know to be continuous on its support. As we did before, we can re-express the resolvent as an ensemble average w.r.t. to the eigenvalue j.p.d.f. $\mathfrak{p}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ and the p.d.f. $\mathcal{P}(X)$ on $\mathcal{S}$:

$$W_1(x) = \left\langle \sum_{i=1}^{n} \frac{1}{x - \lambda_i} \right\rangle = \left\langle \mathrm{Tr} \frac{1}{x - \hat{X}} \right\rangle_{\mathcal{P}(\hat{X})} \tag{1.1.22}$$

These three expression facilitates the dealing of the eigenvalue densities. In fact, dealing with the resolvent will be many times preferred as it presents an asymptotic expansion the densities does not.

Theorem 1.1.8. *(Borot-Guionnet) Assuming certain hypothesis on the eigenvalues j.p.d.f.* $\mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n)$*, the resolvent* $W_1(x)$ *and, more generally, the undefined connected m-point correlator* $W_m(x_1, x_2, \ldots, x_m)$*, have large n expansions of the form*

$$W_m(x_1, x_2, \ldots, x_m, n) = n^{2-m} \sum_{l=0}^{\infty} \frac{W_m^l(x_1, x_2, \ldots, x_m)}{n^l}, \tag{1.1.23}$$

where the correlator expansion coefficients $W_m^l(x_1, x_2, \ldots, x_m)$ *have no dependence on n.*

Some of the conditions of the theorem above guide us to rescale the eigenvalues of certain ensembles so that they are contained in a compact connected set in the large limit $n \to \infty$ regime. In these cases, we study the *global scaled eigenvalue density* $\tilde{\mathtt{p}}(\lambda)$ and the *scaled resolvent* $\tilde{W}_1(x)$ given by

$$\tilde{\mathtt{p}}(\lambda) := \frac{c_n}{n} \mathtt{p}(c_n \lambda) \tag{1.1.24}$$

$$\tilde{W}_1(x) := c_n W_1(c_n x) = n \int_{\mathrm{supp}(\tilde{\mathtt{p}})} \frac{\tilde{\mathtt{p}}(\lambda)}{x - \lambda} \mathrm{d}\lambda \quad \text{for } x \in \mathbb{C} \setminus \mathrm{supp}(\tilde{\mathtt{p}}). \tag{1.1.25}$$

where c_n is a scaling parameter that differs from ensemble from ensemble. Now, assuming the theorem holds, one have that $\tilde{W}_1 = \sum_{l=1}^{\infty} W_1^l(x) n^{1-l}$. Using the Sokohotski-Plemelj inversion formula we define

$$\mathtt{p}^l(\lambda) := \lim_{\epsilon \to 0} \frac{W_1^l(\lambda - \mathrm{i}\epsilon) - W_1^l(\lambda + \mathrm{i}\epsilon)}{2\pi \mathrm{i}}. \tag{1.1.26}$$

Then, $\tilde{\mathtt{p}}(\lambda)$ and $\sum_{l=0}^{\infty} \mathtt{p}^l(\lambda) n^{-l}$ do not agree at finite n, as one might expect. Instead, the latter sum is referred to as the *smoothed eigenvalue density*, as it lacks the oscillatory terms that are known to be present in the true scaled eigenvalue density $\tilde{\mathtt{p}}(\lambda)$.

1.1.3 Classical Ensembles

The denomination *Classical Ensemble* classically denotes the Gaussian, Laguerre and Jacobi ensembles because of their - future discussed - relation to the orthogonal polynomials. We can define them as below.

Definition 1.1.9. *We define the following classical ensembles*

- *Let* $\hat{G}$ *be drawn from the* $n \times n$ *real, complex or quaternionic Ginibre Ensemble. Then, the random matrix* $\hat{H} = \frac{1}{2}(\hat{G}^\dagger + \hat{G})$ *represents the corresponding Gaussian ensemble (also known as Hermite).*
- *Let* $\hat{G}$ *be drawn from the* $n \times m$ *real, complex or quaternionic Ginibre Ensemble. Then, the random matrix* $\hat{W} = (\hat{G}^\dagger \hat{G})$ *represents the corresponding* (m, n) *Laguerre ensemble (also known as Wishart).*
- *We say* $\hat{Y}$ *an element of the* (m_1, m_2, n) *real, complex and quaternionic Jacobi Ensemble if it is of the form* $\hat{Y} = \hat{W}_1(\hat{W}_1 + \hat{W}_2)^{-1}$ *where each* $\hat{W}_i$ *belong to the corresponding* (m_i, n) *Laguerre ensembles with* $m_i \geq n$.

Note that all ensembles presented are self-adjoint and therefore have all n eigenvalues. Gaussian ensembles are also examples of Wigner Matrices.

Proposition 1.1.10. *Let* $k := \beta/2$, $a = k(m_1 - n + 1) - 1$ *and* $b = k(m_2 - n + 1) - 1$*. From the p.d.f. of the Ginibre ensembles, we can change variables to obtain the p.d.f.s of the* $n \times n$ *Gaussian,* (m_1, n) *Laguerre, and* (m_1, m_2, n) *Jacobi ensembles, respectively*

$$\mathtt{p}^{(G)}(\hat{H}) = \frac{1}{\mathcal{Z}_{n,\beta}^{(G)}} \exp\left(-\operatorname{Tr}(\hat{H}^2)\right) \tag{1.1.27}$$

$$\mathtt{p}^{(L)}(\hat{W}) = \frac{1}{\mathcal{Z}_{n,a}^{(L)}} (\det(\hat{W}))^a \exp\left(-\operatorname{Tr}(\hat{W})\right) \tag{1.1.28}$$

$$\mathtt{p}^{(J)}(\hat{Y}) = \frac{1}{\mathcal{Z}_{n,a,b}^{(J)}} (\det(\hat{Y}))^a (\det(\hat{I}_n - \hat{Y})) \tag{1.1.29}$$

With $\beta = 1, 2, 4$ corresponding to the choice of $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathrm{He}$, the random matrix ensembles introduced in the definition are defined with the latter p.d.f,s onto the respective matrix sets

$$\begin{aligned}\mathcal{S}^{(G)} &= \{\hat{H} \in M_{n,n}(\mathbb{F}) | \hat{H} = \hat{H}^{\dagger}\} \\ \mathcal{S}^{(L)} &= \{\hat{W} \in \mathcal{S}^{(G)} | \hat{W} \quad \text{positive definite}\} \\ \mathcal{S}^{(J)} &= \{\hat{Y} \in \mathcal{S}^{(L)} | \hat{I}_n - \hat{Y} \quad \text{positive definite}\}\end{aligned} \tag{1.1.30}$$

The contemporary definition of classical ensembles introduces another ensemble into the family.

Definition 1.1.11. *A random matrix ensemble is said to be a classical matrix ensemble if its eigenvalue j.p.d.f. is of the form*

$$\mathtt{p}^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_n; \beta) = \frac{1}{\mathcal{N}^{(w)}_{n,\beta}} \prod_{i=1}^{n} w(\lambda_i) |\Delta_n(\lambda)|^{\beta}, \quad \beta = 1, 2, 4, \tag{1.1.31}$$

where $\mathcal{N}^{(w)}_{n,\beta}$ is the normalization constant, $\Delta_n(\lambda)$ is the Vandermonde determinant and the weight function $w(\lambda)$ is such that

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \log(w(\lambda)) = \frac{w'(\lambda)}{w(\lambda)} = \frac{f(\lambda)}{g(\lambda)} \tag{1.1.32}$$

with f and g polynomials of degree at most one and two, respectively.

Up to fractional linear transformations $\lambda \mapsto (a_{11}\lambda + a_{12})/(a_{21}\lambda + a_{22})$ with $a_{ij} \in GL_2(\mathbb{R})$, there are precisely four weight functions that satisfies the condition. These are

$$w(\lambda) = \begin{cases} \mathrm{e}^{-\lambda^2} & \text{Gaussian,} \\ \lambda^a \, \mathrm{e}^{-\lambda} \, \chi_{\{\lambda > 0\}} & \text{Laguerre,} \\ \lambda^a (1-\lambda)^b \chi_{\{0<\lambda<1\}} & \text{Jacobi,} \\ (1 - \mathrm{i}\lambda)^{\eta} (1 + \mathrm{i}\lambda)^{\overline{\eta}} & \text{Generalized Cauchy.} \end{cases} \tag{1.1.33}$$

where χ is the indicator function.

Definition 1.1.12. *Taking $k = \beta/2$ and $c_n = \sqrt{kn}, kn, 1$ respectively for the Gaussian, Laguerre and Jacobi ensembles. The global scaled eigenvalue densities of the classical matrix ensembles are*

$$\begin{aligned}\tilde{\mathfrak{p}}^{(G)}(\lambda) &= \sqrt{\frac{k}{n}} \mathfrak{p}^{(G)}(\sqrt{kn}\lambda) \\ \tilde{\mathfrak{p}}^{(L)}(\lambda) &= k \mathfrak{p}^{(L)}(kn\lambda) \\ \tilde{\mathfrak{p}}^{(J)}(\lambda) &= \frac{1}{n} \mathfrak{p}^{(J)}(\lambda) \\ \tilde{\mathfrak{p}}^{(Cy)}(\lambda) &= \frac{1}{n} \mathfrak{p}^{(J)}(\lambda)\end{aligned} \tag{1.1.34}$$

1.2 Invariant Measure

Invariant ensembles of random matrices are probability spaces which are invariant under conjugation of the appropriate matrix group - these ensembles only depend on their eigenvalue distributions while their eigenvectors become independent and are Haar distributed. To properly understand this concept, let's make precise the definition of three infamous ensembles: Unitary, Orthogonal and Symplectic ensembles. Other ensembles many times can be of interest, but their theory may be less straightforward.

1. *Orthogonal Ensembles* consist of $n \times n$ real symmetric matrices $\hat{M} = \overline{\hat{M}} = \hat{M}^T$ together with a probability distribution that is invariant under orthogonal conjugation $\hat{M} \mapsto \hat{O}\hat{M}\hat{O}^T$, $\hat{O}$ orthogonal, i.e., $\hat{O}\hat{O}^T = \hat{O}^T\hat{O} = \hat{I}$.
2. *Unitary Ensembles* consists of $n \times n$ Hermitian matrices $\hat{M} = \hat{M}^*$ together with a probability distribution that is invariant under unitary conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}^*$, $\hat{U}$ unitary, i.e., $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$.

3. *Sympĺectic Ensembles* consists of $2m \times 2m$ Hermitian self-dual matrices $\hat{M} = \hat{M}^* = \hat{J}\hat{M}^T\hat{J}^T$ where $\hat{J} = \operatorname{diag}(\sigma, \cdots, \sigma)$ with

$$\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

together with a probability distribution that is invariant under unitary-symplectic conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}$, with $\hat{U}$ unitary and symplectic, i.e. $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$ and $\hat{U}\hat{J}\hat{U}^T = \hat{J}$.

Examples of such invariant ensembles are the Gaussian Orthogonal, Unitary and Sympĺectic ensembles - *GOE*, *GUE*, *GSE*. These take Gaussian distributed and independent entries. We may also choose to include the Ginibre Ensembles in the types of invariance we will work with. The three types of Ginibre Emsembles: Ginibre Orthogonal, Unitary and Sympĺectic are matrices with independent, identically distributed standard complex Gaussian entries. In this case, the eigenvalues are distributed in the complex plane.

Let's make it precise. Given a group $\mathcal{G}$ and a group action $\mathcal{A} : \mathcal{G} \times \mathcal{S} \to \mathcal{S}$, a random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is said to be $\mathcal{A}$-invariant if the probability measure $\mathrm{d}\mathcal{P}(\hat{X}) := \mathcal{P}(\hat{X})\mathrm{d}\hat{X}$ satisfies

$$\mathrm{d}\mathcal{P}(\mathcal{A}(g, \hat{X})) = \mathrm{d}\mathcal{P}(\hat{X}) \tag{1.2.1}$$

for every $g \in \mathcal{G}$. The term *invariant matrix ensemble* usually refers to invariance under the mapping $\hat{X} \mapsto g\hat{X}g^{-1}$ for g drawn from the appropriate compact group $\mathcal{G}$; when a matrix ensemble is considered to be irreducible and its underlying set of matrix is $S \subseteq \mathbb{M}_{n,n}(\mathbb{R}), \mathbb{M}_{n,n}(\mathbb{C}), \mathbb{M}_{n,n}(\mathrm{He})$, the group $\mathcal{G}$ must be one of $O(n), U(n), Sp(2n)$. Then, we say the ensemble is orthogonal, unitary or symplectic.

We can also determine the probability measure $\mathrm{d}\mathcal{P}(\hat{X})$ to be invariant under the adjoint action of a group $\mathcal{G}$ is we establish that both the p.d.f. $\mathcal{P}(\hat{X})$ and the Lebesgue measure $\mathrm{d}\hat{X}$ are invariant under this action. We can say directly using 1.1.2 and 1.1.10 that the p.d.f.s for Ginibre, Gaussian, Laguerre and Jacobi are invariant under the appropriate adjoint actions.

Proof. (p.d.f.s for Ginibre, Gaussian, Laguerre and Jacobi are invariant under the appropriate adjoint actions.) □

The invariance of the Lebesgue measure is more involved and depends on whether or not the associated matrix set $\mathcal{S}$ consists of purely of self-adjoint matrices. Because of that, we start with the case of the Gaussian, Laguerre and Jacobi ensembles.

Proposition 1.2.1. *Given an $n \times n$ real ($\beta = 1$), complex ($\beta = 2$) or quaternionic ($\beta = 4$) self-adjoint matrix variable $\hat{X}$, diagonalize it as $\hat{X} = \hat{U}\hat{\Lambda}\hat{U}^\dagger$ where $\hat{\Lambda} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ is a diagonal matrix of eigenvalues, and $\hat{U}$ is corresponding matrix of eigenvectors; the spectral theorem dictates that $\hat{U}$ is an element of $O(n)$ in the real case, $U(n)$ in the complex case, and $Sp(2n)$ in the quaternionic case. To make the mapping $\hat{X} \mapsto \hat{U}\hat{\Lambda}\hat{U}^\dagger$ bijective, we insist that the eigenvalues are listed in increasing order and the first row of $\hat{U}$ is non-negative real. The canonical Lebesgue measure*

$$\mathrm{d}\hat{X} = \prod_{i=1}^{n} \mathrm{d}X_{ii} \prod_{1 \le j < k \le n} \prod_{s=1}^{\beta} \mathrm{d}X_{jk}^{s} \tag{1.2.2}$$

on the set of adjoint matrices decomposes as

$$\mathrm{d}\hat{X} = |\Delta_n|^\beta \mathrm{d}\hat{\Lambda}\mathrm{d}\mu_{Haar}(\hat{U}) \tag{1.2.3}$$

where $\mathrm{d}\hat{\Lambda} = \prod_{i=1}^{n} \mathrm{d}\lambda_i$ is the Lebesgue measure on the Weyl Chamber $\lambda_1 < \cdots < \lambda_n \subset \mathbb{R}^n$ and

$$\mathrm{d}\mu_{Haar}(\hat{U}) = \bigwedge_{1 \le l < k \le n} \bigwedge_{s=1}^{\beta} \left(\hat{U}^\dagger [\mathrm{d}U_{ij}]_{i,j=1}^{n} \right)_{lk}^{(s)} \tag{1.2.4}$$

is the unnormalized Haar measure on the quotient space $O(n)/O(1)^n$ ($\beta = 1$), $U(n)/U(1)^n$ ($\beta = 2$), or $Sp(2n)/Sp(2)^n$ ($\beta = 4$), given by the exterior product of the independent real components of the entries of the matrix of differentials $\hat{U}^\dagger[\mathrm{d}U_{ij}]_{i,j=1}^{n}$.

At this point, one can check (albeit with some non-trivial computations) that the Lebesgue measure is indeed invariant in these cases for the appropriate group action.

The situation concerning the $n \times n$ Ginibre Ensemble is more complicated. In the 1965 work, Ginibre extended the proposition above and its arguments starting with the diagonalization formula $\hat{G} = \hat{P}\hat{\Lambda}\hat{P}^{-1}$ - note that, with $\hat{\Lambda}$

diagonal of eigenvalues and $\hat{P}$ of eigenvectors, this formula is valid with probability one since the subset of matrices with repeated eigenvalues has measure zero. However, $\hat{P}$, diagonalizing matrix, is not necessarily orthogonal, unitary, nor symplectic. Nonetheless, QR decomposition can be used to write $\hat{P} = \hat{U}\hat{T}$ where $\hat{U}$ is as before and $\hat{T}$ is an upper triangular matrix with real positive diagonal entries. Then,

$$\mathrm{d}\hat{G} = J(\lambda)\mathrm{d}\hat{\Lambda}\mathrm{d}\mu_{Tri}(\hat{T})\mathrm{d}\mu_{Haar}(\hat{U}) \tag{1.2.5}$$

where $\mathrm{d}\hat{\Lambda}$ is the Lebesgue measure on the space housing the eigenvalues, $\mathrm{d}\mu_{Haar}(\hat{U})$ is the Haar measure defined before, $J(\lambda)$ is a Jacobian consisting of products of Vandermonde determinants involving the eigenvalues, and $\mu_{Tri}(\hat{T})$ is some measure dependent on the entries of T. From here, it follows that the Lebesgue measure on $\hat{G}$ is invariant under the desired conjugation action.

1.2.1 Eigenvalues j.p.d.f.s

To obtain the eigenvalue j.p.d.f. $\mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ from the probability measure $\mathrm{d}\mathcal{P}(\hat{X})$, one changes variable to the eigenvalues and a set of auxiliary variables, then integrates over the latter. This will be something like this sketch

$$\begin{aligned}
\mathcal{P}(\hat{X})\mathrm{d}\hat{X} &= \mathcal{P}(X_{1,1}, \cdots, X_{n,n}) \prod_{i,j} \mathrm{d}M_{i,j} \\
&= \mathcal{P}\left(X_{1,1}(\hat{\Lambda}, \hat{U}), \cdots, X_{n,n}(\hat{\Lambda}, \hat{U})|J(\hat{X} \mapsto \{\hat{\Lambda}, \hat{U}\})\right) \mathrm{d}\hat{\Lambda}\mathrm{d}\hat{U}. \\
\implies \mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n) &= \int_{\mathcal{G}(n)/\mathcal{G}(1)^n} \mathcal{P}(X_{1,1}(\hat{\Lambda}, \hat{U}), \cdots, X_{n,n}(\hat{\Lambda}, \hat{U})|J(\hat{X} \mapsto \{\hat{\Lambda}, \hat{U}\}))\mathrm{d}\hat{\Lambda}\mathrm{d}\hat{U} \\
&\overset{invariance}{=} \mathcal{P}(X_{1,1}(\hat{\Lambda}), \cdots, X_{n,n}(\hat{\Lambda})|J(\hat{X} \mapsto \{\hat{\Lambda}\}))\mathrm{d}\hat{\Lambda} \int_{\mathcal{G}(n)/\mathcal{G}(1)^n} \mathrm{d}\hat{U} \\
&= \mathcal{P}(X_{1,1}(\hat{\Lambda}), \cdots, X_{n,n}(\hat{\Lambda}))|\Delta_n(\lambda)|^\beta \, \mathrm{Vol}(\mathcal{G}(n)/\mathcal{G}(1)^n)\mathrm{d}\hat{\Lambda}
\end{aligned} \tag{1.2.6}$$

A nice observation is that we know $\mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ to be a multivariate symmetric function - this implies that $\mathcal{P}(X_{1,1}(\hat{\Lambda}), \cdots, X_{n,n}(\hat{\Lambda})) \coloneqq \mathcal{P}(\hat{\Lambda})$ to be a function of $\{\mathrm{Tr}(\hat{\Lambda}^k)\}_{k=0}^n$, hence $\{\mathrm{Tr}(\hat{X}^k)\}_{k=0}^n$. For the self-adjoint cases, these auxiliary variables were the quotient spaces of orthogonal, unitary and symplectic matrices. For Ginibre, is the upper triangular entries. Leaving the details of Ginibre behind, we focus on the Gaussian, Laguerre and Jacobi ensembles. Set $\mathcal{P}(\hat{X})$ the p.d.f. for one of the ensembles, fix $(\beta, \mathcal{G}(n)) = (1, O(n)), (2, U(n)), (4, Sp(2n))$. The propositions cited implies that

$$\int_{\mathcal{G}(n)/\mathcal{G}(1)^n} \mathrm{d}\mathcal{P}(\hat{X}) = \mathcal{P}(\Lambda)|\Delta_n(\lambda)|^\beta \mathrm{d}\hat{\Lambda} \int_{\mathcal{G}(n)/\mathcal{G}(1)^n} \mathrm{d}\mu_{Haar}(\hat{U}) \tag{1.2.7}$$

As the eigenvalue and eigenvector statistics are decoupled, one reads off the eigenvalue j.p.d.f. is thus given by

$$\mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n) = \frac{\mathrm{Vol}(\mathcal{G}(n)/\mathcal{G}(1)^n)}{N!} \mathcal{P}(\hat{\Lambda})|\Delta_n(\lambda)|^\beta, \tag{1.2.8}$$

where, of course,

$$\mathrm{Vol}(\mathcal{G}(n)/\mathcal{G}(1)^n) = \int_{\mathcal{G}(n)/\mathcal{G}(1)^n} \mathrm{d}\mu_{Haar}(\hat{U}). \tag{1.2.9}$$

Now, with $w(\lambda)$ the appropriate classical weight, we can check that

$$\mathcal{P}(\hat{\Lambda}) = \prod_{i=1}^n w(\lambda_i),$$

which confirms the consistence with the definition of the eigenvalue j.p.d.f. on the classical ensembles. Furthermore, note that

$$\mathcal{Z}_{n,\beta} = \frac{\mathrm{Vol}(\mathcal{G}(n)/\mathcal{G}(1)^n)}{N!} \mathcal{N}_{n,\beta} \tag{1.2.10}$$

as long as the volume is known.

1.2.2 Gaussian Ensembles

Let's make it explicit the probability distribution on Gaussian Ensembles. We define $\hat{M} = n^{-\frac{1}{2}}\hat{M}' \in \mathcal{M}_{n,n}(\mathcal{F})$ to be of the Gaussian Ensemble if the entries of $\hat{M}'$ are independent and distributed as follows:

$$M'_{i,j} \sim \begin{cases} \mathcal{N}_{\mathcal{F}}(0,1/2) & \text{for } i \neq j, \\ \mathcal{N}_{\mathcal{F}}(0,1) & \text{for } i = j, \end{cases} \quad \text{if } i \leq j.$$

where $\mathcal{F}$ is such as the real line, complex plane or quaternion space. Note that this turns $\hat{M}$ into a Scaled Wigner Matrix. Now, take for example $\mathcal{F} = \mathbb{R}$ and

$$\mathcal{P}(\hat{M}) = \prod_{i=1}^{n} \frac{\exp\left(-\frac{M_{i,i}^2}{2}\right)}{\sqrt{2\pi}} \prod_{i<j}^{n} \frac{\exp\left(-M_{i,i}^2\right)}{\sqrt{\pi}} = 2^{-n/2}\pi^{-n(n+1)/4} \exp\left(-\frac{1}{2}\ \mathrm{Tr}\,\hat{M}^2\right). \tag{1.2.11}$$

Then, we know the distribution of eigenvectors and eigenvalues is the product of the distribution of entries times the Jacobian of the diagonalization, that is

$$\mathcal{P}(\hat{M}) = \mathcal{P}(\hat{\Lambda}, \hat{O}) \prod_{i<j} |\lambda_j - \lambda_i| \tag{1.2.12}$$

Now, for the distribution on the eigenvalues, we integrate on $\mathbb{V}_n$ as

$$\mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n) = \int_{\mathbb{V}_n} \mathcal{P}(\hat{\Lambda}) \prod_{i<j} |\lambda_j - \lambda_i| \mathrm{d}\hat{O} = \mathcal{P}(\hat{\Lambda}) \prod_{i<j} |\lambda_j - \lambda_i| \,\mathrm{Vol}(\mathbb{V}_n) \tag{1.2.13}$$

which lead us to write

$$\mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n) = \frac{1}{\mathcal{Z}_{n,\beta=1}} \exp\left\{-\frac{1}{2}\sum_{i=1}^{n} \lambda_i^2\right\} \prod_{i<j}^{n} |\lambda_i - \lambda_j|. \tag{1.2.14}$$

with $\mathcal{Z}_{n,\beta}$ the partition function, or normalizer of the expression. We could also do the same calculation for the GUE and GSE ensembles, which would result in a similar expression

$$\begin{aligned} \mathtt{p}(\lambda_1, \lambda_2, \ldots, \lambda_n) &= \frac{1}{n!\mathcal{Z}_{n,\beta}^{(ord)}} \exp\left\{-\left(\sum_{i=1}^{n} \frac{\lambda_i^2}{2} - \sum_{i<j}^{n} \log|\lambda_i - \lambda_j|^{\beta}\right)\right\}, \\ &= \frac{1}{\mathcal{Z}_{n,\beta}^{(ord)}}\, \mathrm{e}^{-\beta_n \mathcal{H}_n(\vec{\lambda})}, \end{aligned} \tag{1.2.15}$$

a very familiar expression for a physicist, as it is a Gibbs measure.

1.2.3 Empirical Measure

In many application one is usually concerned with the limiting behavior of the spectra as the size of the matrix tends to infinity, either to properly understand large matrices or to approximate infinite-dimensional operators. A usual tool to describe the eigenvalues of a random matrix is the empirical measure: If $\hat{M}$ has eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, then its empirical spectral measure μ_M is the random probability measure putting equal mass at each of the eigenvalues:

$$\mu_M = \frac{1}{n}\sum_{i=1}^{n} \delta_{\lambda_i} \tag{1.2.16}$$

We want to say that this measure converges, in some sense, for some other measure. A common way of doing this is the concept of *convergence in expectation.* Define $\mathbb{E}(\mu_M)$ to be as follows: for a test function f, define

$$\int f \mathrm{d}\mathbb{E}(\mu_M) := \mathbb{E}\left(\int f \mathrm{d}\mu_M\right) \tag{1.2.17}$$

If we have a sequence of matrices $\{\hat{M}_n\}$ we say that their spectral measure converges in expectation to a limiting measure μ if the sequence $\{\mathbb{E}(\mu_{M_n})\}$ converges weakly to μ, that is, if

$$\int f \mathrm{d}\mathbb{E}(\mu_{M_n}) \overset{n\to\infty}{\to} \int f \mathrm{d}\mu \tag{1.2.18}$$

for bounded continuous test functions f. A natural class of test functions are polynomials. Note that if $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the random eigenvalues of $\hat{M}$ random matrix, we have that

$$\int (z^k + c_{k-1}z^{k-1} + \cdots + c_0)\mathrm{d}\mu_M = \frac{1}{n}\sum_{i=1}^{n}(\lambda_i^k + c_{k-1}\lambda_i^{k-1} + \cdots + c_0) = \frac{1}{n}\, Tr(V(\hat{M})) \tag{1.2.19}$$

In a way, the potential somehow influences the convergence.

1.3 Orthogonal Polynomial Ensembles

The subclass of invariant matrix ensembles this section refers to are those self-adjoint matrices $\hat{X}$ with p.d.f.s $\mathcal{P}(\hat{X})$ such that one has the decomposition

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^{n} w(\lambda_i), \tag{1.3.1}$$

where $\{\lambda_i\}$ are the undistinguishable eigenvalues of $\hat{X}$ and $w(\lambda)$ is some continuous weight function that is real valued on $\mathbb{R}$ and decays fast enough as $|\lambda| \to \infty$. This is the same as removing the constrain on the weight function of the classical ensembles.

Definition 1.3.1. *Let $\beta = 1, 2, 4$, fix $n \in \mathbb{N}$ and let $w(\lambda)$ be a continuous, non-negative, real-valued function such that $w(\lambda) = o(\lambda^{-\beta(n-1)-1})$. Then, the j.p.d.f.*

$$\mathtt{p}^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_n; \beta) = \frac{1}{\mathcal{N}_{n,\beta}^{(w)}} \prod_{i=1}^{n} w(\lambda_i)|\Delta_n(\lambda)|^{\beta} \tag{1.3.2}$$

*describes the n-sized orthogonal polynomial ensemble (OPE) when $\beta = 2$, skew-orthogonal polynomial ensemble of real type ($\mathbb{R}$-SOPE) when $\beta = 1$, and the skew-orthogonal polynomial ensemble of quaternion type (*He*-SOPE) when $\beta = 4$.*

The condition on the weight function guarantees that the j.p.d.f has a well-defined normalization constant. To properly understand the nomenclature used, one must discuss the Vandermonde determinant. Check 1.3.1 for this.

If one wishes to integrate the determinantal and Pfaffian structures given by the Jacobian against the product of weight functions, it is beneficial to use monic polynomials $\{q_j(\lambda)\}_{j=0}^{\infty}$ that are skew-orthogonal with respect to the weight $w(\lambda)$.

Definition 1.3.2. *With $w(\lambda)$ a suitable function, define the following inner products on polynomial space:*

$$\langle f, g\rangle^{(1)} := \frac{1}{2}\int_{\mathbb{R}^2} f(x)g(y)\,\mathrm{sign}(y-x)w(x)w(y)\mathrm{d}x\mathrm{d}y \tag{1.3.3}$$

$$\langle f, g\rangle^{(2)} := \int_{\mathbb{R}} f(x)g(x)w(x)\mathrm{d}x \tag{1.3.4}$$

$$\langle f, g\rangle^{(4)} := \frac{1}{2}\int_{\mathbb{R}} (f(x)g'(x) - f'(x)g(x))w(x)\mathrm{d}x \tag{1.3.5}$$

For $\beta = 1, 2, 4$, let $\{q_j^{(\beta)}\}_{j=0}^{\infty}$ be a family of monic polynomials such that each $q_j^{(\beta)}(\lambda)$ is of degree j. Impose also that exists finite positive constants $\{r_j^{(\beta)}\}_{j=0}^{\infty}$ s.t.

$$\langle q_j^{(2)}, q_k^{(2)}\rangle^{(2)} = r_j^{(2)}\chi_{j=k} \tag{1.3.6}$$

$$\langle q_{2j}^{(\sigma)}, q_{2k+1}^{(\sigma)}\rangle^{(\sigma)} = -\langle q_{2k+1}^{(\sigma)}, q_{2j}^{(\sigma)}\rangle^{(\sigma)} = r_j^{(\sigma)}\chi_{j=k} \tag{1.3.7}$$

$$\langle q_{2j}^{(\sigma)}, q_{2k}^{(\sigma)}\rangle^{(\sigma)} = \langle q_{2j+1}^{(\sigma)}, q_{2k+1}^{(\sigma)}\rangle^{(\sigma)} = 0 \tag{1.3.8}$$

for $\sigma = 1, 4$ and any $j, k > 0$. Then, we say that the family $\{q_j^{(1)}(\lambda)\}_{j=0}^{\infty}$ if $\mathbb{R}$-skew-orthogonal, $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ is orthogonal, and $\{q_j^{(4)}(\lambda)\}_{j=0}^{\infty}$ is He*-skew-orthogonal. All in respect to $w(\lambda)$.*

We can now express the eigenvalue j.p.d.f. $\mathtt{p}^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_n; \beta)$ as either a determinant or a Pfaffian.

Theorem 1.3.3. *Let $\{q_j^{(\beta)}(x)\}_{j=0}^{\infty}, \{r_j^{(\beta)}\}_{j=0}^{\infty}$, be as in the definition above, and let $n \in \mathbb{N}$ with n even in the case for $\beta = 1$. Next, introduce the auxiliary functions*

$$\phi_j(x) := \int_{\mathbb{R}} q_j^{(1)}(y)\,\mathrm{sign}(x-y)w(y)\mathrm{d}y \tag{1.3.9}$$

$$S_j^{(1)}(x,y) := \phi'_{2j+1}(x)\phi_{2j}(y) - \phi'_{2j}(x)\phi_{2j+1}(y) \tag{1.3.10}$$

$$S_j^{(4)}(x,y) := q'_{2j+1}(x)q_{2j}(y) - q'_{2j}(x)q_{2j+1}(y) \tag{1.3.11}$$

$$D_j^{(4)}(x,y) := q'_{2j+1}(x)q'_{2j}(y) - q'_{2j}(x)q'_{2j+1}(y) \tag{1.3.12}$$

$$I_j^{(4)}(x,y) := q_{2j+1}(x)q_{2j}(y) - q_{2j}(x)q_{2j+1}(y) \tag{1.3.13}$$

and the kernels corresponding respectively to the real, complex and quaternionic cases,

$$K_n^{(1)}(x,y) := \sum_{k=0}^{N/2-1} \frac{1}{r_k^{(1)}} \begin{bmatrix} \frac{1}{2}\int_{\mathbb{R}} \mathrm{sign}(x-z)S_k^{(1)}(z,y) & S_k^{(1)}(y,x) \\ -S_k^{(1)}(x,y) & \delta_y S_k^{(1)}(x,y) \end{bmatrix} - \frac{1}{2}\begin{bmatrix} \mathrm{sign}(x-y) & 0 \\ 0 & 0 \end{bmatrix}, \tag{1.3.14}$$

$$K_n^{(2)}(x,y) := \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{r_k^{(2)}} q_k^{(2)}(x) q_k^{(2)}(y), \tag{1.3.15}$$

$$K_n^{(4)}(x,y) := \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{r_k^{(4)}} \begin{bmatrix} I_k^{(4)}(x,y) & S_k^{(4)}(y,x) \\ -S_k^{(4)}(x,y) & D_k^{(4)}(x,y) \end{bmatrix}. \tag{1.3.16}$$

Then, the eigenvalue j.p.d.f. of Definition 1.3.1 given by

$$p^{(w)}(\lambda_1, \lambda_2, \ldots, \lambda_n; \beta) = \begin{cases} \det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}, & \beta = 2, \\ \mathrm{Pf}\left[K_n^{(\beta)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}, & \beta = 1,4. \end{cases} \tag{1.3.17}$$

The significance of this theorem is that, for $\beta = 1,2,4$ the kernels $K_n^{(\beta)}(x,y)$ have *Reproducing properties* such as

$$\int_{\mathbb{R}} \det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k+1} \mathrm{d}\lambda_{k+1} = (n-k)\det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k} \tag{1.3.18}$$

$$\int_{\mathbb{R}} \mathrm{Pf}\left[K_n^{(\beta)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k+1} \mathrm{d}\lambda_{k+1} = (n-k)\,\mathrm{Pf}\left[K_n^{(\beta)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k} \tag{1.3.19}$$

Thus, one may integrate the j.p.d.f. over $\mathrm{d}\lambda_{k+1}\cdots\mathrm{d}\lambda_n$ to obtain the *k-point correlation function*

$$\mathtt{p}_k^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_k;n;\beta) := \frac{n!}{(n-k)!}\int_{\mathbb{R}^{n-k}} \mathtt{p}^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_n;\beta)\mathrm{d}\lambda_{k+1}\cdots\mathrm{d}\lambda_n = \begin{cases} \det\left[K_n^{(2)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}, & \beta = 2, \\ \mathrm{Pf}\left[K_n^{(\beta)}(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}, & \beta = 1,4. \end{cases} \tag{1.3.20}$$

This implies, that he eigenvalue density $p^{(w)} = p_1^{(w)}(\lambda)$ is given by the formula

$$p^{(w)}(\lambda;\beta) = \begin{cases} K_n^{(2)}(\lambda,\lambda), & \beta = 2, \\ \mathrm{Pf}\left[K_n^{(\beta)}(\lambda,\lambda)\right], & \beta = 1,4. \end{cases} \tag{1.3.21}$$

1.3.1 The case for $\beta = 2$

We start illustrating the results above for the case $\beta = 1$. Begin with the following characterization of the Jacobian as a Vandermonde determinant illustrated in 1.3.1

$$\prod_{i<j}^{n} |\lambda_i - \lambda_j|^2 = \det(\lambda_i^{j-1})^2{}_{i,j=1,\cdots,n}. \tag{1.3.22}$$

Now, based on the property 1.4.1, we can rewrite the entries using some class of appropriate polynomial. For this case, we will use $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ orthogonal polynomials. Then,

$$\prod_{i<j}^{n}|\lambda_i-\lambda_j|^2=\det\left(q_{j-1}^{(2)}(\lambda_i)\right)_{n\times n}^2 \tag{1.3.23}$$

Using the fact that $(\det(A))^2=\det(\sum_{j=1}^{n}A_{ji}A_{jk})$ we can rewrite

$$\prod_{i<j}^{n}|\lambda_i-\lambda_j|^2=\det\left(\sum_{j=1}^{n}q_{j-1}^{(2)}(\lambda_i)q_{j-1}^{(2)}(\lambda_k)\right)_{n\times n} \tag{1.3.24}$$

Now, remember that

$$\mathtt{p}^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_n;2)\propto\prod_{i=1}^{n}w(\lambda_i)|\Delta_n(\lambda)|^2, \tag{1.3.25}$$

then we will put the weight inside the determinant in the following manner: Take $w(\lambda_i)=\sqrt{w(\lambda_i)}\sqrt{w(\lambda_i)}$ and multiply one term by the i-th column and the other by the i-th line of the matrix. Now, we should get

$$\mathtt{p}^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_n;2)\propto\det\left(\sqrt{w(\lambda_i)w(\lambda_k)}\sum_{j=1}^{n}q_{j-1}^{(2)}(\lambda_i)q_{j-1}^{(2)}(\lambda_k)\right)_{i,k=1}^{n} \tag{1.3.26}$$

and therefore

$$\mathtt{p}^{(w)}(\lambda_1,\lambda_2,\ldots,\lambda_n;2)\propto\det\left[K_n^{(2)}(\lambda_i,\lambda_k)\right]_{i,k=1}^{n}. \tag{1.3.27}$$

The general formula can also be further simplified but this computation is easier with $\beta=2$, so that's the case we will deal with for now. First, it is known that the orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ satisfy the three-term recurrence

$$q_j^{(2)}(\lambda)=(\lambda+a_j)q_{j-1}^{(2)}(\lambda)-\frac{r_{j-1}^{(2)}}{r_{j-2}^{(2)}}q_{j-2}^{(2)}(\lambda) \tag{1.3.28}$$

where the coefficients depend on the choice of weight. We use this recurrence to write the terms in the summation of $K_n^{(2)}$ so that it gives rise to the *Christoffel-Darboux* formula:

Proposition 1.3.4. *The $\beta=2$ kernel defined before simplifies as*

$$K_n^{(2)}(x,y)=\frac{\sqrt{w(x)w(y)}}{r_{n-1}^{(2)}}\frac{q_n^{(2)}(x)q_{n-1}^{(2)}(y)-q_{n-1}^{(2)}(x)q_n^{(2)}(y)}{x-y} \tag{1.3.29}$$

Taking the limit $y\to x$ using L'Hopital's rule shows that for finite n the density has the form

$$\mathfrak{p}^{(w)}(\lambda)=K_n^{(2)}(\lambda,\lambda)=\frac{\sqrt{w(x)w(y)}}{r_{n-1}^{(2)}}\left(q_{n-1}^{(2)}(\lambda)\partial_\lambda q_n^{(2)}(\lambda)-q_n^{(2)}(\lambda)\partial_\lambda q_{n-1}^{(2)}(\lambda)\right) \tag{1.3.30}$$

All that remains to make $\mathfrak{p}^{(w)}(\lambda)$ fully explicit is knowledge of the orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$. In the Gaussian, Laguerre and Jacobi cases, there are Hermite, Laguerre and Jacobi orthogonal polynomials, respectively.

1.4 Vandermonde

We have said before that the change of variables performed for the invariant ensembles give rise to an Jacobian of the form

$$\prod_{i<j}^{n}|\lambda_j-\lambda_i|^\beta,$$

lets try to properly understand this object. The first thing to note is that this is a determinant. This is done in two steps: First, we show that for a given matrix form, the determinant has exactly this form, secondly we show that the Jacobian is exactly this matrix. For the first part, consider a matrix V given by

$$\hat{V} = \begin{bmatrix} 1 & \alpha_1 & \alpha_1^2 & \dots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \dots & \alpha_2^{n-1} \\ 1 & \alpha_3 & \alpha_3^2 & \dots & \alpha_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \dots & \alpha_n^{n-1} \end{bmatrix}$$

We can use of the properties of the determinant, namely that it does not change if we add a multiple of one row to another row to write

$$|\hat{V}| = \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & \alpha_2 - \alpha_1 & \alpha_2(\alpha_2 - \alpha_1) & \dots & \alpha_2^{n-2}(\alpha_2 - \alpha_1) \\ 1 & \alpha_3 - \alpha_1 & \alpha_3(\alpha_3 - \alpha_1) & \dots & \alpha_3^{n-2}(\alpha_3 - \alpha_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n - \alpha_1 & \alpha_n(\alpha_n - \alpha_1) & \dots & \alpha_n^{n-2}(\alpha_n - \alpha_1) \end{vmatrix} = \prod_{i=2}^{n}(\alpha_i - \alpha_1) \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 1 & 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}$$

Using now of Laplace Theorem we now that there is only one cofactor that matters. So we write

$$|\hat{V}| = \prod_{i=2}^{n}(\alpha_i - \alpha_1) \begin{vmatrix} 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}$$

Repeating this many times we of course have the expression we started with, that is

$$|\hat{V}| = \prod_{i<j}^{n} |\alpha_j - \alpha_i|.$$

We can now use of the properties of the Vandermonde determinant to rewrite the Jacobian as we need. For reasons that will become clearer, we will use the following properties

Afirmation 1.4.1. *Let $\hat{V}$ be a matrix with entries $\{V_{i,j}\} = v_i^{j-1}$ such that its determinant is given by $\prod_{i<j}^{n} |v_j - v_i|$. Then, if we define $\hat{V}'$ with $\{V_{i,j}\} = p(v_i)$ with p a monomial with degree j.*

Chapter 2

Orthogonal Polynomials

Orthogonal Polynomials are of great importance for the theory of random matrices, as their properties allow us to perform operations that would otherwise be very convoluted - see (1.3.19) for example. We start our discussion with $\beta = 2$ - Hermite Polynomials when we use the Gaussian weight function, for its relatively simplicity.

2.1 Hermite Polynomials

Consider $q_n^{(2)}$ when $w^{(G)}(x) := (\sqrt{2\pi})^{-1} \mathrm{e}^{-\frac{1}{2}x^2}$. We make standard the notation

$$\mathrm{He}_n(x) := q_n^{(2,w^{(G)})}. \tag{2.1.1}$$

This means that *Hermite Polynomials* are such that

$$\int_{-\infty}^{\infty} \mathrm{He}_n(x)\mathrm{He}_m(x) w^{(G)}(x)\mathrm{d}x = \begin{cases} 0, & \text{if} n \neq m \\ n!, & \text{if} n = m. \end{cases} \tag{2.1.2}$$

There are some other representation for the polynomial - one of the most common is their derivative formula

$$\mathrm{He}_n(x) = (-1)^{(n)} \mathrm{e}^{\frac{1}{2}x^2} \frac{\mathrm{d}}{\mathrm{d}x^n} \mathrm{e}^{-\frac{1}{2}x^2} \tag{2.1.3}$$

or even the recursion formula

$$\mathrm{He}_{n+1}(x) = x\mathrm{He}_n(x) - n\mathrm{He}_{n-1}(x). \tag{2.1.4}$$

One can also write the explicit representation (or *Standart*),

$$\mathrm{He}_n(x) = n! \sum_{l=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^l \frac{(x)^{n-2l}}{2^l (n-2l)! l!}. \tag{2.1.5}$$

This is all good, but can be a little inconvenient when doing computation. A nice representation is presented by [1]. The Gaussian integral formula for the Hermite Polynomials. This development will be based on this reference.

Definition 2.1.1. *The Gaussian integral Formula, or equivalently the Gaussian Expectation formula, for the Hermite polynomials* He_n *is given by*

$$\mathrm{He}_n(x) = \mathbb{E}_{Z\sim\mathcal{N}(0,1)}[(x+\mathrm{i}Z)^n] = \int_{-\infty}^{\infty} (x+\mathrm{i}z)^n \mathrm{e}^{-\frac{1}{2}z^2} \mathrm{d}z \tag{2.1.6}$$

This deeply simplifies some computation. For example, note that someone can easily compute the identities

$$\mathrm{He}_n(0) = \mathbb{E}[(0+\mathrm{i}Z)^n] = \begin{cases} (-1)^{n/2}(n-1)!!, & \text{for } n \text{ even} \\ 0, & \text{for } n \text{ odd} \end{cases}, \tag{2.1.7}$$

$$\mathrm{He}_n(x+y) = \mathbb{E}[(x+y+\mathrm{i}Z)^n] = \mathbb{E}\left[\sum_{k=0}^{n} \binom{n}{k} y^k (x+\mathrm{i}Z)^{n-k}\right] = \sum_{k=0}^{n} \binom{n}{k} y^k \mathrm{He}_{n-k}(x). \tag{2.1.8}$$

And with that one can get a explicit expression for the polynomial setting $x = 0$ in the second expression and using the first one to calculate the values.

2.1.1 Arbitrary Variance

We can use this formula to work with Hermite Polynomials of arbitrary variance. Assume one adds a temporal dependence parameter t. Denote this polynomial $\tilde{\mathrm{H}}_n(x;t)$ defined as

Definition 2.1.2.

$$\tilde{\mathrm{H}}_n(x;t) := \mathbb{E}_{B_t\sim\mathcal{N}(0,t)}\left[(x+\mathrm{i}B_t)^n\right] = \mathbb{E}_{Z\sim\mathcal{N}(0,1)}\left[(x+\mathrm{i}\sqrt{t}Z)^n\right] \tag{2.1.9}$$

The notation has the tilde because historically there is a preference for the derivative formula instead of the integral formula. The notation H_n is reserved for the dual of this polynomial, defined as below

Definition 2.1.3.

$$\mathrm{H}_n(x;t) := \mathbb{E}_{Z\sim\mathcal{N}(0,\frac{1}{t})}\left[(\frac{x}{t}+\mathrm{i}Z)^n\right] = \mathbb{E}_{Z\sim\mathcal{N}(0,1)}\left[(\frac{x}{t}+\mathrm{i}\frac{Z}{\sqrt{t}})^n\right]. \tag{2.1.10}$$

Which is dual in the sense that

$$\int_{-\infty}^{\infty}\tilde{\mathrm{H}}_n(x;t)\mathrm{H}_m(x;t)(\sqrt{2\pi t})^{-1}\,\mathrm{e}^{-\frac{1}{2}\frac{x^2}{t}}\,\mathrm{d}x = \begin{cases}0, & \mathit{if} n\neq m\\ n!, & \mathit{if} n=m.\end{cases} \tag{2.1.11}$$

Finally, note that

$$\tilde{\mathrm{H}}_n(x;t) = t^{n/2}\tilde{\mathrm{H}}\left(\frac{x}{\sqrt{t}};1\right) = t^{n/2}\mathrm{He}\left(\frac{x}{\sqrt{t}}\right),\quad \mathrm{H}_n(x;t) = t^{-n/2}\tilde{\mathrm{H}}\left(\frac{x}{\sqrt{t}};1\right) = t^{-n/2}\mathrm{He}\left(\frac{x}{\sqrt{t}}\right). \tag{2.1.12}$$

2.1.2 Combinatorial Arguments

Our attention will be on the Asymptotic of the Hermite Polynomials, to get there, we will work our way with some combinatorial arguments. We start with the *Wick Formula/Isserlis' Theorem* for Gaussian expectations.

Theorem 2.1.4. *(Wick Formula/Isserlis' Theorem) If $Z_1, Z_2, \ldots, Z_n$ are a collection of mean zero, jointly Gaussian random variables, then the expectation of the product $Z_1\cdots Z_n$ can be written in terms of the individual correlations $\mathbb{E}[Z_aZ_b]$ in the following way*

$$\mathbb{E}[Z_1\cdots Z_n] = \sum_{\pi\in\mathrm{P}(\{1,\cdots,n\})}\prod_{\{a,b\}\in\pi}\mathbb{E}[Z_aZ_b] \tag{2.1.13}$$

where $\mathrm{P}(\{1,\cdots,n\})$ are the set of pairings (or pair partitions) on the set $S=\{1,\cdots,n\}$. That is, $\mathrm{P}(S)$ denotes the way of dividing S into disjoint pairs.

If one applies this result for example, for $\mathbb{E}[Z^n]$, the expectation of a n-repeated Gaussian Variable, this result gives the moments of the normal distribution

$$\mathbb{E}[Z^n] = |\mathrm{P}(\{1,\cdots,n\})|\cdot\mathbb{E}[Z^2]^{n/2} = \begin{cases}(n-1)!!\cdot\mathbb{V}ar[Z]^{n/2}, & \text{if } n \text{ is even,}\\ 0, & \text{if } n \text{ is odd.}\end{cases} \tag{2.1.14}$$

If we want to apply this combinatorial argument for the polynomials $\tilde{\mathrm{H}}_n(x;t) = \mathbb{E}_{Z\sim\mathcal{N}(0,1)}[(x+\mathrm{i}\sqrt{t}Z)^n]$, we must expand the product in one of the terms on each of the brackets - which yields a sum over 2^n monomials. Then, taking the expectation over Z amounts to adding up the pairings for each term as in 2.1.4. The number of pairings depends on the number of times m that $\mathrm{i}\sqrt{t}Z$ was chosen: there are $(m-1)!!$ such pairings if m is even and 0 pairings if m is odd. Therefore, the only contributions come from the terms with even order.

When exactly m factors $\mathrm{i}\sqrt{t}Z$ are chosen, there must be $n-m$ copies of x chosen, and so the resulting contribution of the term is $(\mathrm{i}\sqrt{t})^m x^{n-m}$. Since m is even we will actually have $(-t)^{m/2}x^{n-m}$. We can count $\binom{n}{m}(m-1)!!$ such terms - m choices of the term and the subsequent $(m-1)!!$ ways of pairing it up. This can be formalized as

$$\begin{aligned}\tilde{\mathrm{H}}_n(x;t) &= \sum_{S\subset\{1,\cdots,n\},\pi\in\mathrm{P}(S)}(\mathrm{i}\sqrt{t})^{|\pi|}x^{n-|S|} && (2.1.15)\\ &= \sum_{\{m=0|m \text{ is even}\}}\binom{n}{m}(m-1)!!(-t)^{m/2}x^{n-m} && (2.1.16)\\ &= \sum_{k=0}^{\lfloor\frac{n}{2}\rfloor}\binom{n}{2k}(2k-1)!!(-t)^k x^{n-2k} && (2.1.17)\\ &= x^n - \binom{n}{2}1!!\cdot tx^{n-2} + \binom{n}{4}3!!\cdot t^2x^{n-4} - \cdots && (2.1.18)\end{aligned}$$

This property can also be used to prove the orthogonality and recursion of the polynomials - for the proof, check [1].

2.1.3 Asymptotic - $x = \mathcal{O}(n)$

We can study the asymptotic of this polynomial for $x = \mathcal{O}(n)$ only using the combinatorial argument given in this chapter.

Proposition 2.1.5. *Let $c \in \mathbb{R}$ be fixed, and set $x = cn$. We have the following asymptotic formula as $n \to \infty$:*

$$\mathrm{He}_n(cn) = (cn)^n \, \mathrm{e}^{-\frac{1}{2c^2}} (1 + o(1)) \tag{2.1.19}$$

that is,

$$\mathrm{He}_n(cn) \approx x^n \, \mathrm{e}^{-\frac{n^2}{2x^2}} \tag{2.1.20}$$

for $x = \mathcal{O}(n)$.

Proof. Using the integral formula 2.1.1 we can write

$$\mathrm{He}_n(cn) = \mathbb{E}[(cn + \mathrm{i}Z)^n] = (cn)^n \mathbb{E}\left[\left(1 + \mathrm{i}\frac{Z}{cn}\right)^n\right]. \tag{2.1.21}$$

Now, note that

$$\lim_{n\to\infty} \left(1 + \mathrm{i}\frac{Z}{cn}\right)^n = \mathrm{e}^{\mathrm{i}\frac{Z}{c}} \tag{2.1.22}$$

almost surely in Z, and in light of the expectation $\mathbb{E}[\mathrm{e}^{\beta Z}] = \mathrm{e}^{\frac{1}{2}\beta^2}$ we find that

$$\mathbb{E}\left[\left(1 + \mathrm{i}\frac{Z}{cn}\right)^n\right] = \mathrm{e}^{-\frac{1}{2c^2}} (1 + o(1)) \tag{2.1.23}$$

from where the results follows. $\square$

Other regimes for the asymptotic will have to use some more sophisticated techniques.

2.1.4 Integral Representation

Follow [4, Proposition 2.1]. A standard method of asymptotic of orthogonal polynomials is the so called *Deift-Zhou* steepest descent method, which considers to asymptotically solve a Riemann-Hilbert problem associated to the orthogonal polynomials. However, the asymptotic may me simpler to compute if the polynomial has an integral representation by direct inspection. The Hermite polynomials have such representation.

Proposition 2.1.6. *The Hermite polynomial* $\mathrm{He}_n(x)$ *has the following ntegral representation*

$$\mathrm{He}_n(x) = \frac{n!}{2\pi\mathrm{i}} \int_{|t|=1} \frac{\mathrm{e}^{tx - \frac{t^2}{2}}}{t^{n+1}} \mathrm{d}t \tag{2.1.24}$$

Proof. We will explicitly arrive at the *Standart* notation for the polynomial using the integral representation

$$\mathrm{He}_n(x) = \frac{n!}{2\pi\mathrm{i}} \int_{|t|=1} \frac{\mathrm{e}^{tx - \frac{t^2}{2}}}{t^{n+1}} \mathrm{d}t \tag{2.1.25}$$

Take the Taylor expansion for the exponential

$$\mathrm{e}^{tx} = \sum_{i=0}^{\infty} \frac{(x)^i}{i!} t^i \tag{2.1.26}$$

$$\mathrm{e}^{-\frac{t^2}{2}} = \sum_{j=0}^{\infty} \frac{(-1)^j}{2^j j!} t^{2j} \tag{2.1.27}$$

$$\tag{2.1.28}$$

and explicitly substitute the expressions into the integral form

$$\mathrm{He}_n(x) = \frac{n!}{2\pi\mathrm{i}} \sum_{i=0}^{\infty} \frac{(x)^i}{i!} \sum_{j=0}^{\infty} \frac{(-1)^j}{2^j j!} \int_{|t|=1} t^{-n-1+i+2j} \mathrm{d}t \tag{2.1.29}$$

and note that this integral is valued zero except when $2j - (i - n) = 0$. Then, we can write, for every even $i - n$,

$$\mathrm{He}_n(x) = n! \sum_{i=0|i-n\text{even}}^{\infty} (-1)^{\frac{n-i}{2}} \frac{x^i}{2^{\frac{n-i}{2}} i! \left(\frac{n-i}{2}\right)!} \tag{2.1.30}$$

$$= n! \sum_{l=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^l \frac{x^{n-2l}}{2^l l! (n-2l)!} \tag{2.1.31}$$

which is exactly the expression given by 2.1.5. □

Proposition 2.1.7. *I believe there is a connection between the two integral representations if we start with the Gaussian expectation formula, expand the binomial and rewrite the moments with the Cauchy formula, as its done to derive this representation starting from the derivative formula. Missing proof.*

Chapter 3

Asymptotic

We have the sense in the previous chapters that we want to study some distributions when our matrices are large enough - be that for feasibility of computations or approximation of infinite dimension operators. This is equivalent, in some sense, to study the regime of large n for some orthogonal polynomials previously discussed. For some of those, a integral representation is available, allowing us to use asymptotic methods on integrals to solve for the large n behavior.

We will try to explore the asymptotic techniques on a specific type of function, as $\lambda \to \infty$, given by

$$\phi(\lambda) := \int_\Gamma g(s)\, \mathrm{e}^{-\lambda f(s)}\, \mathrm{d}s, \tag{3.0.1}$$

where Γ is a contour in the plane and f, g are given suitable functions. Suitable here, means that we impose g and f to be continuous (or analytic), f to have global maximum with zero derivative (maybe in more than one point) and strictly positive second derivative at this point(s) of maximum. There is also a more technical regularity condition on f to ensure the approximations holds.

Motivations

- Airy Equation $\psi''(x) = x\psi(x)$: Taking the Fourrier transform we have

  $$-\omega^2\Phi(x) = \mathrm{i}\Phi'(x)$$

  for which we can solve for $\Phi(x)$ and then take the inverse Fourrier Transform to get

  $$\phi(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}/3\omega^3 + \mathrm{i}x\omega}\, \mathrm{d}\omega.$$

  The imaginary part of the argument is an odd function and therefore zero when integrated. The rest is expressed

  $$\phi(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \cos\left(\frac{\omega^3}{3} + x\omega\right)\mathrm{d}\omega.$$

  But how to know if this is the only solution? One can prove that for a more general definition of a Transform

  $$\tilde{F}f(w) := \frac{1}{\sqrt{2\pi\mathrm{i}}}\int_\Gamma \mathrm{e}^{-\mathrm{i}\omega s}\, f(s)\mathrm{d}s$$

  holds very similar properties of the Fourier Transformation and solves the equation for any contour that gives a convergent integral. We can actually choose independent contour such that each give us a different independent solution to the equation.

3.1 Laplace Method

All approximation in this chapter will use in some way the Lemma of Laplace, or the Laplace Method for integrals of the form

$$\Phi(\lambda) := \int_a^b g(x)\, \mathrm{e}^{-\lambda\phi(x)}\, \mathrm{d}x \tag{3.1.1}$$

We first state a non-precise version of this theorem and discuss some ideas surrounding it.

Theorem 3.1.1. *Let Φ be as in (3.1.1), with $g(x)$ a continuous function and $f(x)$ assuming a global maximum at $x^* \in (a,b)$ with $f'(x^*) = 0$ and $f''(x^*) > 0$. Then, under some regularity conditions, the function Φ satisfies an estimate of the form*

$$\Phi(\lambda) = \mathrm{e}^{-\lambda\phi(x^*)} \frac{\sqrt{2\pi} g(x^*)}{\sqrt{|\phi''(x^*)|}\lambda^{1/2}} (1 + o(1)) \tag{3.1.2}$$

as $\lambda \to \infty$.

The idea of the proof is quite more revealing than the proper result. We abstain for now from giving a proper formal proof and instead give a general sketch that should be enough to illustrate the important point for this section.

Proof. (Sketch of Proof) The main idea is to approximate ϕ around its global maximum by Taylor's theorem

$$\phi(x) = \phi(x^*) - \frac{1}{2}|\phi''(x^*)|(x - x^*)^2 + \epsilon_2(x - x^*) \tag{3.1.3}$$

and then perform a change of variables $\frac{s}{\sqrt{\lambda}} = x - x^*$, so that

$$\int_{x^*-\frac{c}{\sqrt{\lambda}}}^{x^*+\frac{c}{\sqrt{\lambda}}} \mathrm{e}^{-\lambda\phi(x)} \,\mathrm{d}(x) = \mathrm{e}^{-\lambda\phi(x^*)} \int_{-c}^{c} \mathrm{e}^{\frac{1}{2}|\phi''(x^*)|t^2 - \lambda\epsilon_2\left(\frac{t}{\sqrt{\lambda}}\right)} \frac{\mathrm{d}t}{\sqrt{\lambda}}. \tag{3.1.4}$$

Now, using the technical assumption of regularity on the function ϕ, one can argue that the error term ϵ_2 goes to zero sufficiently fast, leaving a Gaussian Integral. More that that, one need to argue that the integral outside this small neighborhood of x^* is exponentially smaller then $\mathrm{e}^{\lambda\phi(x^*)}$ and use continuity of g to include it in the argument. $\square$

Some important points to highlight:

1. More precise error estimates can be obtained if one uses higher order terms of the Taylor Series expansion of ϕ.
2. This argument and result could also be used on curves on the plane. This is clear from the fact that every (smooth) curve is locally a straight line. This will lead us to the Method of *Steepest Descent*.
3. For maximizer x^* which have null second derivative (a double critical point), the Laplace Approximation can be used expanding to the third order terms of the Taylor Series $\phi(x) = \phi(x^*) = \frac{1}{6}\phi'''(x^*)(x - x^*)^3 + \epsilon_3(x - x^*)$ and making the change of variables $\frac{t}{\sqrt[3]{n}} = x - x^*$ instead. This is precisely one of the cases in the expansion for the Hermite Polynomial.

These next subsections are in the spirit of proving a bit more formally the expansion given by the Laplace Method.

3.1.1 Watson's Lemma

We start this study in the real line by looking at the functions $F : \mathbb{R} \to \mathbb{R}$

$$F(\lambda) = \int_0^T f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t \tag{3.1.5}$$

which can be identified as the Laplace transform of a function f supported on $[0, T]$.

Theorem 3.1.2. *(Watson's Lemma) Take $f \in L^1(0,T)$ and suppose that for some $\delta > 0$ it is of the form*

$$f(t) = t^\alpha f_0(t), \quad 0 < t \leq \delta,$$

where $\alpha \in (-1, \infty)$ and f_0 is continuously differentiable on $[0, \delta]$ with $f_0(0) \neq 0$. Then the function F in 3.1.5 satisfies

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha + 1)}{\lambda^{\alpha+1}}(1 + \mathcal{O}(\lambda^{-1})) \tag{3.1.6}$$

when $\lambda \to \infty$ and where Γ is the Gamma function.

Proof. A weak estimative of the integral values would be given by

$$\int_0^T f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t \le ||f||_{L^1}$$

where we only use the fact that $f \in L^1(0,T)$. However, we have more information on the behavior of the function in the section $0 < t < \delta$. To use this information we start by decomposing the function in two parts

$$F(\lambda) = (1) + (2) = \int_0^\delta f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + \int_\delta^T f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t.$$

For (2), we can use the weak estimative, leaving us with

$$F(\lambda) \le \int_0^\delta f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + ||f||_{L^1}\,\mathrm{e}^{-\delta\lambda}.$$

Now, we remind of the condition of continous differentiability and the hypothesis on $f(t)$ to write (1) as

$$(1) = \int_0^\delta t^\alpha (f_0(0) + r(t))\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = f_0(0)\int_0^\delta t^\alpha\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + \int_0^\delta t^\alpha r(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = f_0(0)\cdot(1.1) + (1.2)$$

For these two integrals we can try to simplify the problem by rewritring

$$(1.1) = \int_0^\infty t^\alpha\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t - \int_\delta^\infty t^\alpha\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - \int_\delta^\infty t^\alpha\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t$$

$$|(1.2)| \le ||f'||_\infty \int_0^\delta t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = ||f'||_\infty\left(\int_0^\infty t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t - \int_\delta^\infty t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right) = ||f'||_\infty\left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - \int_\delta^\infty t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right).$$

We are now only missing one ingredient to get the result, the order of

$$G(\lambda,\beta) := \int_\delta^\infty t^\beta\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t, \quad \lambda > 1, \quad \beta > -1.$$

Using Cauchy-Schwartz we can estimate

$$\begin{aligned}
|G(\lambda,\beta)|^2 &\le \left(\int_\delta^\infty \mathrm{e}^{-\lambda t}\,\mathrm{d}t\right)\left(\int_\delta^\infty t^{2\beta}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right) \\
&= \left(\frac{e^{-\lambda\delta}}{\lambda}\right)\left(\int_\delta^\infty t^{2\beta}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right) \\
&\overset{(u=\lambda t)}{=} \left(\frac{e^{-\lambda\delta}}{\lambda}\right)\left(\int_\delta^\infty \mathrm{e}^{-u}\,u^{2\beta}\frac{1}{\lambda^{2\beta+1}}\mathrm{d}u\right) \\
&= \frac{\mathrm{e}^{-\lambda\delta}}{\lambda^{2\beta+2}}M_\beta
\end{aligned}$$

Where M_β is independent of λ. That leave us with

$$G(\lambda,\beta) = \mathcal{O}\left(\frac{\mathrm{e}^{-\lambda\delta/2}}{\lambda^{\beta+1}}\right) = \mathcal{O}\left(\mathrm{e}^{-\epsilon\lambda}\right)$$

where $\epsilon > 0$ depends on β and δ. Finally we can rewrite

$$(1.1) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - G(\lambda,\alpha) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + \mathcal{O}\left(\mathrm{e}^{-\epsilon\lambda}\right)$$

$$|(1.2)| \le ||f'||_\infty\left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - G(\lambda,\alpha+1)\right) = ||f'||_\infty\left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} + \mathcal{O}\left(\mathrm{e}^{-\epsilon\lambda}\right)\right) = \mathcal{O}\left(\lambda^{-\alpha-2}\right)$$

and finally

$$F(\lambda) \le \mathcal{O}\left(\lambda^{-\alpha-2}\right) + f_0(0)\left(\frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + \mathcal{O}\left(\mathrm{e}^{-\epsilon\lambda}\right)\right) + ||f||_{L^1}\,\mathrm{e}^{-\delta\lambda}.$$

by the dominance of the exponential error, we can finally get the result

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + \mathcal{O}(\lambda^{-1}))$$

□

In the estimative the Gamma function and the factor of λ^-1 is the result of integrating the linear exponent of e given by $-\lambda t$. If we, however, change this exponent to be of the general type $-\lambda t^d$ we can still have a similar result on the behavior of the function.

Proposition 3.1.3. *For $T, S > 0$, define the function*

$$G(\lambda) := \int_{-S}^{T} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t$$

and assume $g(x)$ absolutely integrable in $[-S, T]$, of class C^1 in a fixed neighborhood of the origin. Then the function satisfies

$$G(\lambda) = \frac{g(0)\Gamma(1/2)}{\lambda^{1/2}} + \mathcal{O}\left(\frac{1}{\lambda^{3/2}}\right) \tag{3.1.7}$$

when $\lambda \to \infty$ and where Γ is the Gamma function.

Proof. The proof is a reduction of the problem to Watson's Lemma. Assume, $T > S$ and write

$$G(\lambda) = \int_{-S}^{S} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t + \int_{S}^{T} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t$$

The second integral we know to be of order

$$\int_{S}^{T} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t \leq ||g||_\infty\,\mathrm{e}^{-\lambda S^2},$$

exponential on λ, so we drop it. For the first integral we rewrite

$$\begin{aligned}\int_{-S}^{S} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t &= \int_{0}^{S} (g(t) + g(-t))\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t \\ &\overset{(u=t^2)}{=} \frac{1}{2}\int_{0}^{S^2} \left(\frac{g\left(\sqrt{u}\right) + g\left(-\sqrt{u}\right)}{\sqrt{u}}\right)\mathrm{e}^{-\lambda u}\,\mathrm{d}u\end{aligned}$$

where now we can use Watson's Lemma. □

3.1.2 Laplace's Lemma

A seemly natural extension of the type of function we were dealing with are functions of the type

$$\Phi(\lambda) := \int_{a}^{b} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \tag{3.1.8}$$

for $a, b \in \mathbb{R} \cup \{-\infty, \infty\}$ and $a < b$. For now, ϕ is a real-valued function. To begin the study let's make some assumptions. The first one is that a is finite, that means that we can set $a = 0$ without loss of generality. The second assumption is that ϕ is a increasing function on $[0, b]$, so that $\phi(0) < \phi(t)$ for any t in the interval. With that we note that the coefficient $\mathrm{e}^{-\lambda\phi(t)} / \mathrm{e}^{-\lambda\phi(0)}$ decreases exponentially, that is, the integral is accumulated as the neighborhood of 0,

$$\Phi(\lambda) \approx \int_{0}^{\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t.$$

As we know $\phi(t)$ to be increasing, it's first non zero derivative must be of even order, so that when we expanded the function by Taylor,

$$\phi(t) \approx \phi(0) + \phi_{2k+1}t^{2k+1}$$

for some $k \geq 0$ and $|t| < \delta$. As ϕ_{2k+1}, the $(2k+1)$th-derivative, must be positive we also have that, as $\lambda \to \infty$,

$$\begin{aligned}\Phi(\lambda) &\approx \mathrm{e}^{-\lambda\phi(0)} \int_{0}^{\delta} g(t)\,\mathrm{e}^{-\lambda\phi_{2k+1}(t)\cdot t^{2k+1}}\,\mathrm{d}t \\ &\overset{(u=t(\lambda\phi_{2k+1})^{1/(2k+1)})}{\approx} \mathrm{e}^{-\lambda\phi(0)}\,\frac{g(0)}{(\lambda\phi_{2k+1})^{\frac{1}{2k+1}}} \int_{0}^{\infty} \mathrm{e}^{-u^{2k+1}}\,\mathrm{d}u \\ &\overset{(s=u^{2k+1})}{\approx} \mathrm{e}^{-\lambda\phi(0)}\,\frac{g(0)}{(2k+1)(\lambda\phi_{2k+1})^{\frac{1}{2k+1}}} \int_{0}^{\infty} \mathrm{e}^{-s}\,\mathrm{d}u\end{aligned}$$

Where the integral can be evaluated as an Gamma function. More than anything we need to focus on an important assumption that was made: that ϕ has a global minima at one of the endpoints of integration. Is that were not the case, for example if $a < 0 < b$ and $\phi(t)$ had a global minima at $t = 0$ we would now have to consider the expansion

$$\phi(t) \approx \phi(0) + \phi_{2k} t^{2k}$$

which would lead us to the expression

$$\Phi(\lambda) \approx \mathrm{e}^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{2k})^{\frac{1}{2k}}} \int_{-\infty}^{\infty} \mathrm{e}^{-u^{2k}} \, \mathrm{d}u$$

which differentiates of the previously result more importantly by the limits of integration. This is closely related to physical boundaries of the problem.

Theorem 3.1.4. *(Laplace's Method, endpoint contributions) Let Φ be as in 3.1.8, with a finite and $g \in L^1(a,b)$. Assume that ϕ has a unique global minimum at $t = a$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form*

$$g(t) = (t-a)^{\alpha} g_0(t)$$

for $|t-a| < \delta$ and g_0 continuously differentiable in $[a, a+\delta]$ and $g_0(a) \neq 0$, whereas ϕ is twice continuously differentiable on $[a, a+\delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \frac{1}{\lambda^{\alpha+1}} \frac{g_0(a)\Gamma(\alpha+1)}{\phi'(a)^{\alpha+1}} \mathrm{e}^{-\lambda\phi(a)} \left(1 + \mathcal{O}(\lambda^{-1})\right) \tag{3.1.9}$$

as $\lambda \to \infty$.

Proof. Start by assuming, w.l.g, that $a = 0$ and dividing the integral in two, one in the region of interest and one in the rest of the interval

$$\Phi(\lambda) = \int_0^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = \int_0^{\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_{\delta}^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = (1) + (2).$$

Knowing that ϕ has a global minima at $t = 0$ we can assume, changing δ as needed, that $\phi(t) > \phi(\delta)$ for any $t > \delta$. That give us a simple limitation for (2):

$$(2) = \int_{\delta}^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \le \mathrm{e}^{-\lambda\phi(\delta)} \int_{\delta}^b g(t)\mathrm{d}t \le \mathrm{e}^{-\lambda\phi(\delta)}\, ||g||_{L^1[0,b]}.$$

We now get back to (1) and start by studying the difference give by $\phi(t) - \phi(0) = s$ in a similar way that we would think of a change of variable. Define the multivariate function

$$H(t,s) := \phi(t) - \phi(0) - s$$

such that $H(0,0) = 0$ and $\partial_t H(0,0) \neq 0$ so that we can use the Theorem of Implicit Function to say that there exists a unique bijective and continuously differentiable function $t(s) := t$ such that

$$\begin{cases} \phi(t(s)) - \phi(0) - s = 0 \\ t'(0) = \frac{\partial_s H(0,0)}{\partial_t H(0,0)} = \frac{1}{\phi'(0)} \end{cases}$$

If we set $T = t^{-1}(\delta)$ we would then have

$$\int_0^{\delta} g(s)\,\mathrm{e}^{-\lambda\phi(s)}\,\mathrm{d}s = \int_0^T g(t(s))\,\mathrm{e}^{-\lambda\phi(t(s))}\,t'(s)\mathrm{d}s$$

for now to apply the assumption on g. We write $g(t(s)) = t^{\alpha}(s) g_0(t(s))$ and because of Taylor,

$$t(s) = st'(0)(1 + r(s)) = \frac{s}{\phi'(0)}(1 + r(s)).$$

Joining both results,

$$\begin{aligned}
\int_0^T g(t(s))\,\mathrm{e}^{-\lambda\phi(t(s))}\,t'(s)\mathrm{d}s &= \mathrm{e}^{\lambda\phi(0)}\int_0^T g(t(s))\,\mathrm{e}^{-\lambda(\phi(t(s))-\phi(0))}\,t'(s)\mathrm{d}s \\
&= \mathrm{e}^{\lambda\phi(0)}\int_0^T t^\alpha(s)g_0(t(s))\,\mathrm{e}^{-\lambda s}\,t'(s)\mathrm{d}s \\
&= \frac{\mathrm{e}^{\lambda\phi(0)}}{\phi'(0)^\alpha}\int_0^T s^\alpha(1+r(s))^\alpha g_0(t(s))\,\mathrm{e}^{-\lambda s}\,t'(s)\mathrm{d}s
\end{aligned} \tag{3.1.10}$$

where, if we set $f(s) := s^\alpha(1+r(s))^\alpha g_0(t(s))$ we can apply Watson's Lemma. □

Theorem 3.1.5. *(Laplace's Method, interior contributions) Let Φ be as in 3.1.8, with a finite and $g \in L^1(a,b)$. Assume that ϕ has a unique global minimum at $t = c \in (a,b)$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form*

$$g(t) = (t-a)^\alpha g_0(t)$$

for $|t-a| < \delta$ and g_0 continuously differentiable in $[c-\delta, c+\delta]$ and $g_0(a) \neq 0$, whereas ϕ is three times continuously differentiable on $[c-\delta, c+\delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \mathrm{e}^{-\lambda\phi(c)}\left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}\lambda^{1/2}} + \mathcal{O}(\lambda^{-3/2})\right) \tag{3.1.11}$$

as $\lambda \to \infty$.

Proof. Although the proof may be similar there is an important distinction in the solution. Suppose we wanted to try the same approach. We would first assume, w.l.g., that c = 0. Then, we could write

$$\Phi(\lambda) = \int_a^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = \int_a^{-\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_{-\delta}^{\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_\delta^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = (1)+(2)+(3).$$

For (1) and (3) the result is as it was before

$$\begin{cases} (1) = \int_\delta^a g(-t)\,\mathrm{e}^{-\lambda\phi(-t)}\,\mathrm{d}t \leq \mathrm{e}^{\lambda\phi(-\delta)}\,||g||_{L^1[a,b]} \\ (3) = \int_\delta^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \leq \mathrm{e}^{\lambda\phi(\delta)}\,||g||_{L^1[a,b]} \end{cases}$$

For (2) we would try to define a function

$$H(t,s) := \phi(t) - \phi(0) - s^2.$$

However we could not apply the Implicit Function Theorem here as its derivative $\partial_t H = \phi'$ vanishes at $t = 0$. A nice fact about this indetermination is that it doesn't come from the fact that there are no solutions to $H(t,s) = 0$ but from the fact that there are two. To solve this we will use a technique called *blow up*. First, we introduce a variable $v = v(t)$ such that $vs = t$. Now, we redefine our function of interested to be

$$L(s,v) := \frac{\phi(sv) - \phi(0)}{s^2} - 1.$$

for $s \in (-\delta, \delta) - \{0\}$. We would think this function is also singular but note that, using the Taylor Expansion of ϕ we can write

$$\phi(t) = \frac{\phi''(0)}{2}t^2 + t^3 R(t)$$

which leave us with

$$L(s,v) = \frac{\phi''(0)}{2}v^2 + sv^3 R(sv) - 1$$

where we can indeed apply the Implicit Function Theorem. In fact, we know $L \in C^2$ and as we want to solve for $L(s,v) = 0$, we need that the v_0 we take to satisfy $L(0, v_0) = 0$, ie,

$$v_0 = \sqrt{\frac{2}{\phi''(0)}}.$$

Now, we also know that

$$\partial_v L(0, v_0) = \phi''(0) v_0 \neq 0.$$

So we get a bijective and continuously differentiable function $v = v(s)$ such that

$$\begin{cases} L(s, v(s)) = 0 \\ v(0) = v_0 \end{cases}$$

But, $L(s, v(s))$ means exactly that

$$\phi(t) - \phi(0) = s^2$$

with $t = t(s) = sv(s)$. Getting back to our equation, we have that

$$\mathrm{e}^{\lambda\phi(0)} \int_{-\delta}^{\delta} g(t)\,\mathrm{e}^{-\lambda(\phi(t)-\phi(0))}\,\mathrm{d}t = \mathrm{e}^{\lambda\phi(0)} \int_{\alpha}^{\beta} g(sv(s))(v(s) + sv'(s))\,\mathrm{e}^{-\lambda s^2}\,\mathrm{d}t$$

where we can easily calculate $\alpha = -\sqrt{\phi(-\delta) - \phi(0)}$ and $\beta = \sqrt{\phi(\delta) - \phi(0)}$. We finish the prove with Proposition 3.1.3. □

Example - Gamma Function

Take the function called Gamma given by the expression

$$\Gamma(x) = \int_0^\infty \mathrm{e}^{-t}\, t^{x-1} \mathrm{d}t \tag{3.1.12}$$

We are interested in what happens to the function when we take the variable x to $+\infty$. To see what happens, let's fit it into Proposition 3.1.5. Rewrite equation 3.1.12 as

$$\int_0^\infty \mathrm{e}^{-t + x\log(t) - \log(t)}\,\mathrm{d}t.$$

This lead us to identify ϕ as $-\log(t)$, which would give us $\phi'(t) < 0$, that is, we would have it's minimum at $t = +\infty$. However $\mathrm{e}^{\log(t)} = t^1$ is not integrable near the origin. Something need to change about the identification. Let's introduce a variable to rescale the integral: $xs = t$, s.t.,

$$\Gamma(x) = x^x \int_0^\infty \mathrm{e}^{-\lambda(s - \log(s))}\,\mathrm{e}^{-s}\,\mathrm{d}s.$$

Where we have used $\lambda = x - 1$. Thus we identify $\phi(s) = s - \log(s)$ and $g(s) = \mathrm{e}^{-s}$. in fact, now ϕ has a global minima at $s = 1$ and $\phi''(1) = 1 \neq 0$. Hence

$$\begin{aligned} \Gamma(x) &= \frac{\sqrt{2\pi} x^x\,\mathrm{e}^{-x}}{(x-1)^{(-1/2)}}(1 + \mathcal{O}(x^{-1})) \\ &= \sqrt{2\pi} x^{x-1/2}\,\mathrm{e}^{-x}(1 + \mathcal{O}(x^{-1})) \\ &= (x - 1/2)\log(x) - x + \frac{1}{2}\log(2\pi) \end{aligned}$$

Which is called the Stirling formula.

3.2 Steepest Descent

Let's now finally explore integrals over the complex plane with complex valued functions. We retake the integral

$$\phi(\lambda) := \int_\Gamma g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}s, \tag{3.2.1}$$

where Γ is a contour in the complex plane and f, g are given *complex-valued, analytic functions.*

Suppose that Γ is parametrized by $\gamma : [a,b] \to \mathbb{C}$ and let us decompose the functions we have into their real and imaginary parts such as

$$f(z) = u(z) + \mathrm{i}v(z), \quad g(z) = p(z) + \mathrm{i}q(z), \quad \gamma(t) = \alpha(t) + \mathrm{i}\beta(t)$$

with all functions in the decomposition being real valued. Of course we now have

$$\mathrm{e}^{-\lambda f(z)} = \mathrm{e}^{-\lambda u(z)}\,\mathrm{e}^{-\mathrm{i}v(z)}$$

and when λ grows large, we have the exponentially growing/decaying term $\mathrm{e}^{-\lambda u(z)}$ and a oscillating term coming from $\mathrm{e}^{\mathrm{i}v(z)}$ which is hard to control. The idea to get around this fact is to use paths in which this term is constant. If we have such a case, we would write

$$\Phi(\lambda) = \mathrm{e}^{-\mathrm{i}\lambda v_0} \int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t + \mathrm{i}\,\mathrm{e}^{-\mathrm{i}\lambda v_0} \int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t \tag{3.2.2}$$

where each integral can now be solved by Laplace's method.

Theorem 3.2.1. *Consider Φ as in 3.2.1 with a smooth contour Γ. Assume that the exponent $f = u + \mathrm{i}v$ is an analytic function on a neighborhood of Γ that satisfies*

$$v(z) \equiv v_0, \quad z \in \Gamma$$

In addition, suppose that u has a unique minimum at an interior point of Γ, say on $z_0 \in \Gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of Γ and satisfies

$$\int_\Gamma |g(z)||dz| < \infty.$$

Then Φ has a leading asymptotic as $\lambda \to +\infty$ given by

$$\Phi(\lambda) = \mathrm{e}^{-\lambda f(z_0)} \left(\frac{\sqrt{2\pi} g(z_0)\,\mathrm{e}^{\mathrm{i}\theta(z_0)}}{\lambda^{1/2}\sqrt{|f''(z_0)|}} + \mathcal{O}(\lambda^{-3/2}) \right), \quad \lambda \to +\infty,$$

where $\theta(z_0)$ is the angle of the oriented tangent of Γ at z_0.

Proof. Fix a parametrization $\gamma : [a,b] \to \mathbb{C}$ of Γ with $\gamma(c) = z_0, a < c < b$, and decompose Φ as before in 3.2.2. By the assumptions on the functions f, g we can apply Laplace‘s method on both equations, and yields, as $\lambda \to +\infty$,

$$\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t = \sqrt{2\pi}\,\mathrm{e}^{-\lambda u(z_0)} \left(\frac{p(z_0)\alpha'(c) - q(z_0)\beta'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + \mathcal{O}\left(\lambda^{3/2}\right) \right)$$

and

$$\int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t = \sqrt{2\pi}\,\mathrm{e}^{-\lambda u(z_0)} \left(\frac{p(z_0)\beta'(c) - q(z_0)\alpha'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + \mathcal{O}\left(\lambda^{3/2}\right) \right)$$

Using that $v_0 + u(z_0) = f(z_0)$ and also that,

$$(p(z_0)\alpha'(c) - q(z_0)\beta'(c)) + \mathrm{i}(p(z_0)\beta'(c) - q(z_0)\alpha'(c)) = g(z_0)\gamma'(c).$$

Combining the estimative for both parts we get

$$\Phi(\lambda) = \sqrt{2\pi}\,\mathrm{e}^{-\lambda f(z_0)} \left(\frac{g(z_0)\gamma'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + \mathcal{O}\left(\lambda^{3/2}\right) \right)$$

and it suffices now to get an expression for the second derivative in the denominator. Knowing the imaginary part is constant we can write

$$\frac{\mathrm{d}}{\mathrm{d}t} u(\gamma(t)) = \frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t)) = \gamma'(t) f'(\gamma(t))$$

and

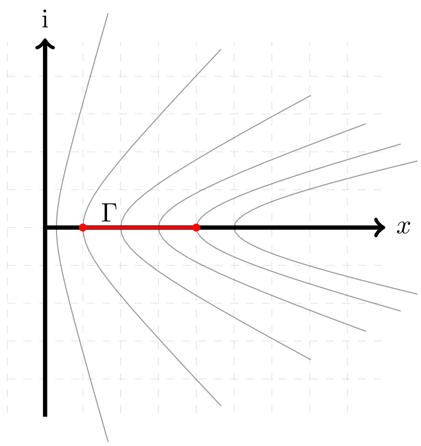
$$\frac{\mathrm{d}^2}{\mathrm{d}t}u(\gamma(t)) = \frac{\mathrm{d}^2}{\mathrm{d}t}f(\gamma(t)) = (\gamma'(t))^2 f''(\gamma(t)) + \gamma''(t)f'(\gamma(t)).$$

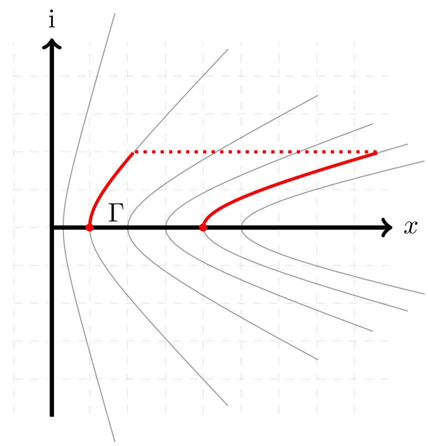
Hence,

$$\frac{\gamma'(c)}{\sqrt{|(u\circ\gamma)''(c)|}} = \frac{1}{\sqrt{|f''(z_0)|}}\frac{\gamma'(c)}{|\gamma'(c)|}$$

which concludes the proof. $\square$

We now need do discuss why this assumption of constant value of $\mathrm{I}m(f(z))$ is not an assumption as strong as one might assume at first. Suppose we want to integrate in a random contour Γ of the plane. Of course, our assumption that the functions were analytic permit us to deform the contour keeping the value of the integral. We are going to use this to find an contour $\tilde{\Gamma}$ such that sections of the contour are such that $\mathrm{I}m(f(z))$ is constant. in the figure above we plot such a scenario by using the level lines of the imaginary part of f as curves on the plane.





The name given by the method is also intuitive. We already know that being analytic means satisfying the *Cauchy-Riemann* equations,

$$u_x(z) = v_y(z), \text{ and } u_y(z) = -v_x(z).$$

An immediate result from these equation is that the gradients of u and v must be normal. Indeed, if $< \cdot, \cdot >$ is the inner product on $\mathbb{R}^2$,

$$< \nabla u(z), \nabla v(z) >= u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

As we're following the curves where $\mathrm{I}m(f(z))$ is constant, we're also following the path of *steepest descent* of the real part of the integral. As we also are looking to use Laplace's method on the integrals, we may want to have our paths going through the critical points of the function.

3.2.1 Airy function

First, we define the function we call Airy.

Definition 3.2.2. *(Airy function) Let $\langle$ be the union of the line segments $\{t\,\mathrm{e}^{\mathrm{i}\pi/3} : t \geq 0\}$ and $\{t\,\mathrm{e}^{-\mathrm{i}\pi/3} : t \geq 0\}$, that is the union of the two infinite rays in the complex plane emanating from the origin that make angles $\pi/3$ and $-\pi/3$ with the real axis. For $u \in \mathbb{C}$ the Airy function is defined by the complex contour integral*

$$Ai(u) = \frac{1}{2\pi\mathrm{i}}\int_{\langle} \exp\left(\frac{t^3}{3} - ut\right)\mathrm{d}t. \tag{3.2.3}$$

Of course, by Cauchy Theorem, we know that the definition is independent of the contour as long as the contour goes from $\mathrm{e}^{-2\pi\mathrm{i}/3}\,\infty$ to $\mathrm{e}^{2\pi\mathrm{i}/3}\,\infty$. Now, we need to get things in order to apply Laplace method on the integral. We could start defining

$$g(z) = \mathrm{e}^{-\frac{1}{3}z^3}, \quad f(z) = z$$

but this is not a good approach as such an f won't have any critical points and won't help us calculate the integral. We then try a change of variable to get a function with those critical points. Define $x := \lambda\,\mathrm{e}^{\mathrm{i}\theta} = \lambda\omega$ so that we can send x to infinite by sending λ in a constant direction θ. Now, introduce a variable s such that $s\lambda^{\eta} = z$ for some $\eta \in \mathbb{R}$. Now, we want to extract λ from the new exponent

$$-\frac{1}{3}\lambda^{3\eta}s^3 + \lambda^{1+\eta}\omega s$$

so we set $\eta = 1/2$ and have that

$$Ai(\lambda,\theta) = \frac{\lambda^{\frac{1}{2}}}{2\pi\mathrm{i}}\int_{\Gamma} \mathrm{e}^{-\lambda^{\frac{3}{2}}(s^3/3-\omega s)}\,\mathrm{d}s$$

$$\implies Ai(\lambda,\theta) = \frac{\lambda^{\frac{1}{2}}}{2\pi\mathrm{i}}\Phi(\lambda^{\frac{3}{2}}), \quad \Phi(\lambda) = \int_{\Gamma} \mathrm{e}^{-\lambda f(s)}\,\mathrm{d}s, \quad f(s) = f_{\omega} := \frac{s^3}{3} - \omega s$$

Note that we have f with two critical points, $s_c^{\pm} = \pm\sqrt{\omega}$. We now need to study the steepest descent path. For reasons that will be clear soon, we divide the analysis into the cases

1. $\theta = 0$
2. $0 < \theta < \frac{2}{3}\pi$
3. $\theta = \frac{2}{3}\pi$
4. $\frac{2}{3}\pi < \theta < \pi$
5. $\theta = \pi$

the rest of the cases should follow from the same arguments based on the symmetry of the critical points. For each case we want to deform Γ to a new contour $\tilde{\Gamma}$ such that

(a) $\tilde{\Gamma}$ is homotopically equivalent to Γ, in the sense that the integral in the definition of $\Phi(\lambda)$ is invariant under the deformation from Γ to $\tilde{\Gamma}$.

(b) The contour $\tilde{\Gamma}$ is a level line of $\mathrm{I}m\,(f)$ that passes through one of the critical points of f.

For (a) to be true we need to guarantee that the deformed path $\tilde{\Gamma}$ should start at $\infty\,\mathrm{e}^{-\mathrm{i}\theta}$ and end at $\infty\,\mathrm{e}^{\mathrm{i}\theta}$ for some $\theta \in [\pi/2, 5\pi/6]$. Now we need to discuss how to satisfy (b) for every one of the cases cited. For all cases, we refer to Image 3.1 for illustration of the scenario and intuition on the contour lines we take.

For the first case, (1), we ca see that the integration path can be taken as the union of the lines emerging from s_c^- with angle $\pi/2$ and $3\pi/2$. For the second case we note that something happens when $\theta = 2/3\pi$. This is because $f(s_c^+) = f(s_c^-)$ when $\theta = 2/3\pi$. But until this point the topology is equivalent. For this interval of values then we take the path of integration to be basically the same as before, the union of the lines emerging from s_c^- with angle $\pi/2$ and $3\pi/2$.

For both of these cases we have the integration of Γ to be given by the contribution of the critical point s_c^- by the asymptotic of Laplace.

For $\theta = 2/3\pi$, case (3), something change. We now need to take a path that passes through both critical points. WE would think that now we have to consider the asymptotic from both saddle points but, in reality, because $Re(f(s_c^-)) < Re(f(s_c^+))$ the contribution from s_c^- exponentially dominates the asymptotic.

For case (4) we no longer have a clear connection of the lines. We need to take $\tilde{\Gamma}$ to be the union of two disjoint curves: The line coming from $\infty\,\mathrm{e}^{-\mathrm{i}2\pi/3}$ passing through s_c^- and going to infinity parallel to the real line and the line coming from infinity parallel to the real line, passing through s_c^+ and going to $\infty\,\mathrm{e}^{\mathrm{i}2\pi/3}$. Again, the contribution from s_c^- exponentially dominates the asymptotic.

For case (5), when $\theta = \pi$, the path of steepest descent is similar to the one just taken. But a key difference is to be noted, now, as $Re(f(s_c^-)) = Re(f(s_c^+))$, contributions from both saddle points must be accounted in the asymptotic.

3.2.2 Hermite's Polynomials

This example will be a collage of many sources: Check [3] and [1] for the main references and notation. We already know the asymptotic for $x = o(n)$, done in 2.1.3. There are other two cases of interest that are related to methods explained in this Chapter. These are $x = o(1)$ and, more importantly, $x = o(\sqrt{n})$.

Regime $x = o(1)$

Proposition 3.2.3. *For $x \in \mathbb{R}$ fixed, we have the following asymptotic as $n \to \infty$:*

$$\mathrm{He}_n(x) = \sqrt{2}\left(\frac{n}{2}\right)^{n/2} \mathrm{e}^{x^2/4} \cos\left(\frac{n\pi}{2} - x\sqrt{n}\right)(1 + o(1)) \tag{3.2.4}$$

Proof. Start with the integral formula given by Property 2.1.1 and write

$$\mathrm{He}_n(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x + \mathrm{i}t)^n \,\mathrm{e}^{-t^2/2}\,\mathrm{d}t = \frac{2}{\sqrt{2\pi}} \Re\left\{\int_0^{\infty} (x + \mathrm{i}t)^n \,\mathrm{e}^{-t^2/2}\,\mathrm{d}t\right\} \tag{3.2.5}$$

$$= \frac{2}{\sqrt{2\pi}} \Re\left\{\int_{\Re(z)=x,\mathrm{Im}(z)>0} \mathrm{e}^{n\ln(z) + \frac{1}{2}(z-x)^2}\, \frac{\mathrm{d}z}{\mathrm{i}}\right\} \tag{3.2.6}$$

$$= \frac{2}{\sqrt{2\pi}} \Re\left\{\int_{\Re(z)=0,\mathrm{Im}(z)>0} \mathrm{e}^{n\ln(z) + \frac{1}{2}(z-x)^2}\, \frac{\mathrm{d}z}{\mathrm{i}}\right\} \tag{3.2.7}$$

where you start with a line integral in $\mathbb{C}$ and move the contour over to the positive imaginary axis - justified by the Cauchy's Theorem and the exponential decay of the integrand near infinity. Now, we focus on $n\ln(z) + \frac{1}{2}(z-x)^2$ which has maximum at $z^* = \mathrm{i}\sqrt{n} + o(\sqrt{n})$. The idea is to change variables by a factor of $\sqrt{z}$ so that the location of the maximum contribution stays $\mathcal{O}(1)$ as n gets large. Specifically, we do

$$z = \mathrm{i}\sqrt{n}s, \quad \frac{\mathrm{d}z}{\mathrm{i}} = \sqrt{n}\mathrm{d}s, \quad \ln(z) = \ln(\sqrt{n}) + \ln(s) + \mathrm{i}\frac{\pi}{2} \tag{3.2.8}$$

which give us

$$\frac{1}{2}(z-x)^2 = -\frac{1}{2}ns^2 + \frac{1}{2}x^2 - \mathrm{i}\sqrt{n}sx \tag{3.2.9}$$

so that we remain with

$$\mathrm{He}_n(x) = \frac{2}{\sqrt{2\pi}} \Re\left\{\int_0^{\infty} \mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{n\ln(s)}\,\mathrm{e}^{n\mathrm{i}\frac{\pi}{2}}\,\mathrm{e}^{\frac{1}{2}x^2}\,\mathrm{e}^{-\mathrm{i}\sqrt{n}sx}\,\sqrt{n}\mathrm{d}s\right\} \tag{3.2.10}$$

$$= \frac{2}{\sqrt{2\pi}}\sqrt{n}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}x^2} \int_0^{\infty} \mathrm{e}^{n\left(\ln(s) - \frac{1}{2}s^2\right)} \cos\left(n\frac{\pi}{2} - \sqrt{n}sx\right)\mathrm{d}s, \tag{3.2.11}$$

where we can apply Laplace's approximation discussed before. Noting that $\phi(s) = \ln(s) - \frac{1}{2}s^2$ has $\phi'(s) = \frac{1}{s} - s$ and $\phi''(s) = -\frac{1}{s^2} - 1$, with global maximum at $s^* = 1$ with $\phi(1) = \frac{1}{2}$ and $\phi''(1) = -2$, yielding

$$\mathrm{He}_n(x) = \frac{2}{\sqrt{2\pi}}\sqrt{n}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}x^2}\left(\mathrm{e}^{n\phi(1)}\sqrt{\frac{2\pi}{n|\phi''(1)|}}\right)\cos\left(n\frac{\pi}{2} - \sqrt{n}x\right)(1 + o(1)), \tag{3.2.12}$$

which simplifies to the result. $\square$

Regime $x = o(\sqrt{n})$

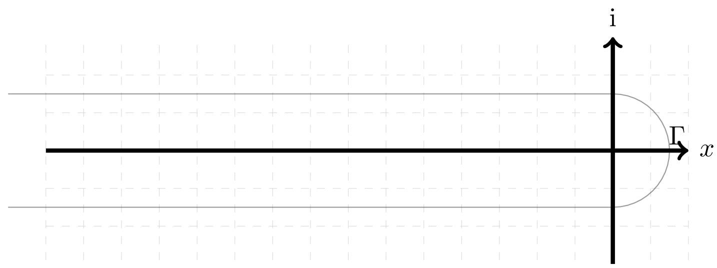
There are three asymptotics regimes for $\mathrm{He}_n(x)$ when $x \to \infty$ depending on the value of $\lim_{N\to\infty} x/2\sqrt{N}$ for $N = n + 1$, each of which has qualitatively different behavior

- Small x values $-2\sqrt{N} < x < 2\sqrt{N}$: in this regime the Hermite Polynomials will be oscillating, and a have a similar qualitative behavior as 3.2.2 with modification to phase and amplitude. These polynomials will have some connections to the *arcsine law* and *semi-circle law*.
- Large x values $|x| > 2\sqrt{N}$: In this regime the polynomial doesn't have more oscillations and is monotone increasing. As x grows larger, this approximation approaches 2.1.3.
- Edge x values $x = \pm 2\sqrt{N} + \mathcal{O}(N^{-\frac{1}{6}})$: in this regime the polynomials will be approximated by the Airy Function. This interpolates between the oscillatory small x regime and the monotone large x regimes.

We write the integral representation given by the Cauchy Integral applied to the Generating Function, this representation should be discussed in 2.1.4 and given by:

$$\begin{aligned}\mathrm{He}_n(x) &:= \frac{n!}{2\pi \mathrm{i}} \int_\Gamma z^{-n-1}\, \mathrm{e}^{xz-\frac{z^2}{2}}\, \mathrm{d}z \\ &= \frac{n!}{2\pi \mathrm{i}} \int_\Gamma \mathrm{e}^{xz-\frac{z^2}{2}-(n+1)\ln(z)}\, \mathrm{d}z\end{aligned} \tag{3.2.13}$$

Where Γ is a contour that goes around the origin coming and going from minus infinity such as $\Gamma = \{\lambda - \mathrm{i}\delta : \lambda < 0\} \cup \{\delta\, \mathrm{e}^{\mathrm{i}\theta} : -\pi/2 < \theta < \pi/2\} \cup \{\lambda + \mathrm{i}\delta : \lambda < 0\}$. This is equivalent to using $\Gamma = \{\mathrm{e}^{\mathrm{i}\theta} : 0 \leq \theta < 2\pi\}$.



There are a few expressions we want to have. Namely we want to determine the formulas for He_n and He_{n-1}. Let's start working on the first case. To do this, set $N = n+1$ and consider the scaling $x \mapsto N^a\alpha$ and $z = N^b\beta$

$$\begin{aligned}\mathrm{He}_n(N^a\alpha) &= \frac{(N-1)!}{2\pi \mathrm{i}} \int_\Gamma \mathrm{e}^{(N^a\alpha)(N^b\beta)-\frac{1}{2}(N^b\beta)^2-N\ln(N^b\beta)}\, \mathrm{d}\beta \\ &\overset{(a=b=1/2)}{=} \frac{(N-1)!N^{\frac{1}{2}(1-N)}}{2\pi \mathrm{i}} \int_\Gamma \mathrm{e}^{-N(-\alpha\beta+\frac{1}{2}\beta^2+\ln(\beta))}\, \mathrm{d}\beta\end{aligned} \tag{3.2.14}$$

So that we have

$$\mathrm{He}_n(\sqrt{N}\alpha) = C_n^0 \int_\Gamma \mathrm{e}^{-N\phi_\alpha(\beta)}\, \mathrm{d}\beta \tag{3.2.15}$$

with

$$\phi_\alpha(\beta) = \frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta). \tag{3.2.16}$$

As we want to calculate the asymptotic by steepest descent, we need to determine the critical points of such a function and then the steepest descent paths passing through this points. We can easily determine the critical points as $\beta_c^\pm = \frac{\alpha\pm\sqrt{\alpha^2-4}}{2}$ - solutions to the quadratic formula given by solving the critical point of the function (the zeros of its derivatives). Now, the behavior of the critical points will depend on the interval of the real parameter α.

$$\beta_c^\pm = \frac{\alpha \pm \sqrt{\alpha^2-4}}{2} = \begin{cases} \mathrm{e}^{\pm\gamma}, & \alpha = 2\cosh(\gamma) > 2, \quad \gamma > 0; \\ \mathrm{e}^{\pm \mathrm{i}\gamma}, & \alpha = 2\cos(\gamma) \in [0,2), \quad \gamma \in (0,\pi/2]; \\ 1, & \alpha = 2. \end{cases} \tag{3.2.17}$$

Therefore, the following theorem follows.

Theorem 3.2.4. *The Hermite Polynomials* $\mathrm{He}_n(x)$ *have the following three representation when* $x = \mathcal{O}(\sqrt{n})$ *and* $x > 0$*:*

1. *Small* x *values* $-2\sqrt{N} < x < 2\sqrt{N}$*: For fixed* $\gamma \in (0,\pi/2)$ *set* $x = 2\sqrt{N}\cos(\gamma)$ *and we obtain*

$$\mathrm{He}_n(2\sqrt{N}\cos(\gamma)) = \sqrt{\frac{2}{\sin(\gamma)}} N^{-\frac{N}{2}}(N-1)^{\left(N-\frac{1}{2}\right)}\mathrm{e}^{N\left(-3\cos^2(\gamma)+\frac{1}{2}\right)+1}\sin\left(\theta(\beta_c^+) - N\left(\frac{\sin(2\gamma)}{2}+\gamma\right)\right)(1+\mathcal{O}(N^{-1})), \tag{3.2.18}$$

$$\mathrm{He}_{n-1}(2\sqrt{N}\cos(\gamma)) = \sqrt{\frac{2}{\sin(\gamma)}} \frac{(N-2)^{N-\frac{3}{2}}}{(N-1)^{\frac{N-1}{2}}} \frac{\mathrm{e}^{N\left(-3\cos^2(\gamma)+\frac{1}{2}\right)+1}}{\mathrm{e}^{\left(3\cos^2(\gamma)-\frac{1}{2}\right)}} \sin\left(\theta(\beta_c^+) - (N-1)\left(\frac{\sin(2\gamma)}{2}+\gamma\right)\right)(1+\mathcal{O}(N^{-1})). \tag{3.2.19}$$

2. *Large x values $|x| > 2\sqrt{N}$: For fixed $\gamma > 1$, write $x = 2\sqrt{N}\cosh(\gamma)$. We obtain*

$$\mathrm{He}_n(2\sqrt{N}\cosh(\gamma)) = \mathrm{e}^{-N\left(\frac{\mathrm{e}^{-2\gamma}}{2}+\gamma+1\right)+1}\,\frac{(N-1)^{N-\frac{1}{2}}N^{-\frac{N}{2}}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}}\left(1+\mathcal{O}(N^{-1})\right), \tag{3.2.20}$$

$$\mathrm{He}_{n-1}(2\sqrt{N}\cosh(\gamma)) = \frac{\mathrm{e}^{-N\left(\frac{\mathrm{e}^{-2\gamma}}{2}+\gamma+1\right)}}{\mathrm{e}^{-\left(\frac{\mathrm{e}^{-2\gamma}}{2}+\gamma-2\right)}}\,\frac{(N-1)^{-\frac{N-1}{2}}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}}\left(1+\mathcal{O}(N^{-1})\right). \tag{3.2.21}$$

3. *Edge x values $x = \pm 2\sqrt{N} + \mathcal{O}(N^{-\frac{1}{6}})$: For fixed $u \in \mathbb{R}$, set $x = 2\sqrt{N} + N^{-1/6}\xi$, then we obtain*

$$\mathrm{He}_n(2\sqrt{N} + N^{-1/6}\xi) = \sqrt{2\pi}(N-1)^{\frac{1}{2}}N^{1/6-N/2}\,\mathrm{e}^{\left(1/2+N^{-1}-2\xi\beta_c N^{-2/3}\right)}\,Ai(\xi)(1+o(1)), \tag{3.2.22}$$

$$\mathrm{He}_{n-1}(2\sqrt{N}+N^{-1/6}\xi) = \sqrt{2\pi}(N-2)^{\frac{1}{2}}(N-1)^{4/6-N/2}\,\mathrm{e}^{\left(1/2+(N-1)^{-1}-2\xi\beta_c(N-1)^{-2/3}\right)}\,Ai(\xi)(1+o(1)). \tag{3.2.23}$$

Case 1

For $-2\sqrt{N} < x < 2\sqrt{N}$, or, equivalently, $-2 < \alpha < 2$ we have, as Figure 3.2 suggests, both critical points as complex values, this can also be easily determined noting that if we set α constant with $0 < \alpha < \sqrt{2}$, we will have

$$\beta_c^{\pm} = \frac{\alpha}{2} \pm \frac{\mathrm{i}}{2}\sqrt{4-\alpha^2}. \tag{3.2.24}$$

And, if we parametrize $\alpha = 2\cos(\gamma)$ for $\gamma \in (0, \pi/2]$ we can rewrite

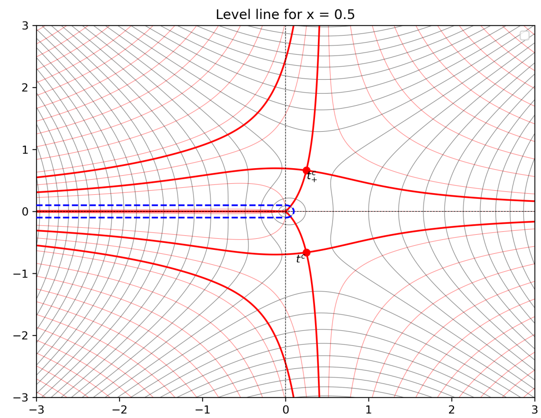
$$\beta_c^{\pm} = \mathrm{e}^{\pm\mathrm{i}\gamma}. \tag{3.2.25}$$

If we are indeed interested in solving the paths where the imaginary parts are constant, as needed for the Laplace method, we will have to solve an equation such as

$$\mathrm{Im}(\phi_\alpha(z) - \phi_\alpha(\beta_c^{\pm})) = 0, \tag{3.2.26}$$

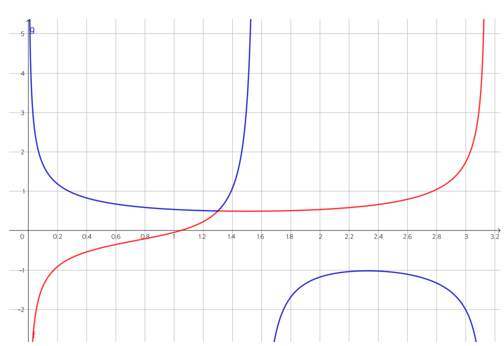
for some $z = r\,\mathrm{e}^{\mathrm{i}\theta}$. We first solve for the positive critical point - the negative will be analogous. This equation can be rewritten as a quadratic formulae

Figure 3.2: Figure for Case 1



$$\frac{1}{2}r^2\sin(2\theta) - r\cos(\gamma)\sin(\theta) + (\theta - \theta_0) = 0 \tag{3.2.27}$$

where we have set $\theta_0 = \gamma - \frac{1}{2}\sin(2\gamma)$. Where we solve for the two solutions r_a^+ and r_d^- by meshing the two quadratic solutions given by the equations. This give us two smooth paths for the positive critical point



$$r_d^+(\theta) = \begin{cases} \frac{\cos(\gamma) - \sqrt{\cos^2(\gamma) - (\theta-\theta_0)/\tan(\theta)}}{\cos(\theta)}; & \theta \in [\gamma, \pi) \\ \frac{\cos(\gamma) + \sqrt{\cos^2(\gamma) - (\theta-\theta_0)/\tan(\theta)}}{\cos(\theta)}; & \theta \in (0, \gamma) \end{cases} \tag{3.2.28}$$

$$r_a^+(\theta) = \begin{cases} \frac{\cos(\gamma) + \sqrt{\cos^2(\gamma) - (\theta-\theta_0)/\tan(\theta)}}{\cos(\theta)}; & \theta \in [\gamma, \pi/2) \\ \frac{\cos(\gamma) - \sqrt{\cos^2(\gamma) - (\theta-\theta_0)/\tan(\theta)}}{\cos(\theta)}; & \theta \in [\theta_0, \gamma) \end{cases} \tag{3.2.29}$$

With which we can construct both paths of constant imaginary part

$$\Gamma_d^+ = \{r_d^+(\theta)\,\mathrm{e}^{\mathrm{i}\theta} : \theta \in (0,\pi)\}; \quad \Gamma_a^+ = \{r_a^+(\theta)\,\mathrm{e}^{\mathrm{i}\theta} : \theta \in (0,\pi/2)\} \tag{3.2.30}$$

Analogously, for the negative part,

$$\Gamma_d^- = \{r_d^-(\theta)\,\mathrm{e}^{\mathrm{i}\theta} : \theta \in (0,\pi)\}; \quad \Gamma_a^- = \{r_a^-(\theta)\,\mathrm{e}^{\mathrm{i}\theta} : \theta \in (0,\pi/2)\} \tag{3.2.31}$$

Now, we only need to determine the steepest descent path. This should be easy as we know the asymptotic behavior $\phi_\alpha(t) \sim t^2$ for $t \to \infty$ and that $\phi_\alpha(t) \sim \ln(t)$ for $t \to 0$. From this, follows that Γ_d^+ and Γ_d^- are the paths we were looking for. We create therefore, a new path Γ_x that connects both Γ_d at the vertical line x. Then, doing $\Gamma \mapsto \Gamma_d^+ \cup \Gamma_d^- \cup \Gamma_x$ and noting that, as x goes to infinity, we have

$$\left|\int_{\Gamma_x} g(x)\,\mathrm{e}^{-n\phi_\alpha(\beta)}\,\mathrm{d}\beta\right| \leq \max_{y\in\Gamma_x}\left|g(x)\,\mathrm{e}^{-n\phi_\alpha(\beta)}\right| \int_{\Gamma_x} |\mathrm{d}\beta| \overset{t\to\infty}{\to} 0 \tag{3.2.32}$$

where $g(t)$ can be both the inverse square root or inverse function. Therefore, we only need to consider the other two paths when considering the asymptotic. Finally, retaking the expression for the Hermite Polynomial,

$$\begin{aligned}
\mathrm{He}_n(2\sqrt{N}\cos(\gamma)) &= C_n^0 \int_\Gamma \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta = C_n^0\left[\int_{\Gamma_d^-} \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta + \int_{\Gamma_d^+} \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta\right] \\
&\overset{(Steepest)}{=} C_n^0\left(\mathrm{e}^{N(\phi_\alpha(\beta_c^+))}\left[\frac{\sqrt{2\pi}}{N^{\frac12}}\frac{\mathrm{e}^{\mathrm{i}(\theta(\beta_c^+))}}{\sqrt{2\sin(\gamma)}} + \mathcal{O}(N^{-\frac32})\right] + \mathrm{e}^{N\overline{(\phi_\alpha(\beta_c^+))}}\left[\frac{\sqrt{2\pi}}{N^{\frac12}}\frac{\mathrm{e}^{\mathrm{i}\overline{(\theta(\beta_c^+))}}}{\sqrt{2\sin(\gamma)}} + \mathcal{O}(N^{-\frac32})\right]\right) \\
&= C_n^0\sqrt{\frac{2\pi}{2N\sin(\gamma)}}\left(\mathrm{e}^{N(\phi_\alpha(\beta_c^+))}\left[\mathrm{e}^{\mathrm{i}(\theta(\beta_c^+))} + \mathcal{O}(N^{-1})\right] + \mathrm{e}^{N\overline{(\phi_\alpha(\beta_c^+))}}\left[\mathrm{e}^{\mathrm{i}\overline{(\theta(\beta_c^+))}} + \mathcal{O}(N^{-1})\right]\right) \\
&= C_n^0\sqrt{\frac{\pi}{N\sin(\gamma)}}\left(\mathrm{e}^{N(\phi_\alpha(\beta_c^+))+\mathrm{i}(\theta(\beta_c^+))} + \mathrm{e}^{N\overline{(\phi_\alpha(\beta_c^+))}+\mathrm{i}\overline{(\theta(\beta_c^+))}}\right)(1+\mathcal{O}(N^{-1})) \\
&= C_n^0\sqrt{\frac{\pi}{N\sin(\gamma)}}\left(\mathrm{e}^{-\mathrm{i}\theta(\beta_c^+)+N(-\alpha\beta_c^+ + \frac12(\beta_c^+)^2+\ln(\beta_c^+))} + \mathrm{e}^{\mathrm{i}\pi}\,\mathrm{e}^{\mathrm{i}\theta(\beta_c^+)+N\overline{(-\alpha\beta_c^+ + \frac12(\beta_c^+)^2+\ln(\beta_c^+))}}\right)(1+\mathcal{O}(N^{-1})) \\
&= C_n^0\sqrt{\frac{\pi}{N\sin(\gamma)}}\,\mathrm{e}^{N(-3\cos^2(\gamma)-\frac12)}\left(\mathrm{e}^{-\mathrm{i}\left(-N\left(\frac{\sin(2\gamma)}{2}+\gamma\right)+\theta(\beta_c^+)\right)} - \mathrm{e}^{\mathrm{i}\left(-N\left(\frac{\sin(2\gamma)}{2}+\gamma\right)+\theta(\beta_c^+)\right)}\right)(1+\mathcal{O}(N^{-1})) \\
&= C_n^0\sqrt{\frac{\pi}{N\sin(\gamma)}}\,\mathrm{e}^{N(-3\cos^2(\gamma)-\frac12)}\,2\mathrm{i}\sin\left(\theta(\beta_c^+) - N\left(\frac{\sin(2\gamma)}{2}+\gamma\right)\right)(1+\mathcal{O}(N^{-1})) \\
&= \frac{(N-1)!N^{\frac12(1-N)}}{2\pi\mathrm{i}}\sqrt{\frac{\pi}{N\sin(\gamma)}}\,\mathrm{e}^{N(-3\cos^2(\gamma)-\frac12)}\,2\mathrm{i}\sin\left(\theta(\beta_c^+) - N\left(\frac{\sin(2\gamma)}{2}+\gamma\right)\right)(1+\mathcal{O}(N^{-1})) \\
&= \sqrt{\frac{2}{\sin(\gamma)}}N^{-\frac{N}{2}}(N-1)^{\left(N-\frac12\right)}\,\mathrm{e}^{N\left(-3\cos^2(\gamma)+\frac12\right)+1}\sin\left(\theta(\beta_c^+) - N\left(\frac{\sin(2\gamma)}{2}+\gamma\right)\right)(1+\mathcal{O}(N^{-1}))
\end{aligned} \tag{3.2.33}$$

Where we used Stirling's Formula and the fact that we have

$$\begin{cases}
\Re[\overline{(-\alpha\beta_c^+ + \frac12(\beta_c^+)^2+\ln(\beta_c^+))}] = -3\cos^2(\gamma) - \frac12; & Im[(-\alpha\beta_c^+ + \frac12(\beta_c^+)^2+\ln(\beta_c^+))] = \frac{\sin(2\gamma)}{2}+\gamma; \\
\Re[\overline{(-\alpha\beta_c^+ + \frac12(\beta_c^+)^2+\ln(\beta_c^+))}] = -3\cos^2(\gamma) - \frac12; & Im[\overline{(-\alpha\beta_c^+ + \frac12(\beta_c^+)^2+\ln(\beta_c^+))}] = -(\frac{\sin(2\gamma)}{2}+\gamma).
\end{cases} \tag{3.2.34}$$

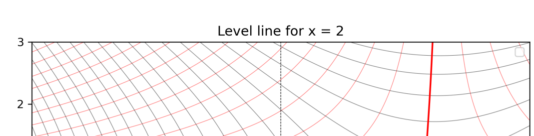
Of course, if we change for the other case we can simply rewrite

$$\mathrm{He}_{n-1}(2\sqrt{N}\cos(\gamma)) = \sqrt{\frac{2}{\sin(\gamma)}}\frac{(N-2)^{N-\frac32}}{(N-1)^{\frac{N-1}{2}}}\frac{\mathrm{e}^{N\left(-3\cos^2(\gamma)+\frac12\right)+1}}{\mathrm{e}^{\left(3\cos^2(\gamma)-\frac12\right)}}\sin\left(\theta(\beta_c^+) - (N-1)\left(\frac{\sin(2\gamma)}{2}+\gamma\right)\right)(1+\mathcal{O}(N^{-1})) \tag{3.2.35}$$

Case 2

Here, with $|x| > 2\sqrt{N}$, or equivalently $\alpha > 2$, we have - as the plot 3.3 suggests - both critical points as real

Figure 3.3: Figure for case 3



values, this can be easily determined noting that if we set $\alpha > 2$, we will have

$$\beta_c^{\pm} = \frac{\alpha}{2} \pm \frac{1}{2}\sqrt{\alpha^2 - 4}. \tag{3.2.36}$$

And, if we parametrize $\alpha = 2\cosh(\gamma)$ for $\gamma > 1$ we can rewrite

$$\beta_c^{\pm} = \mathrm{e}^{\pm\gamma}. \tag{3.2.37}$$

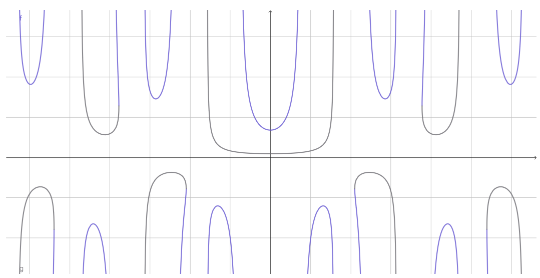
If we are indeed interested in solving the paths where the imaginary parts are constant, as needed for the Laplace method, we will have to sove an equation such as

$$\mathrm{Im}(\phi_\alpha(z) - \phi_\alpha(\beta_c^{\pm})) = 0, \tag{3.2.38}$$

for some $z = r\,\mathrm{e}^{\mathrm{i}\theta}$. We first solve for the positive critical point and he negative will be analogous. This can be rewritten as as quadratic equation

$$\frac{1}{2}r^2\sin(2\theta) - r\cosh(\gamma)\sin(\theta) + \theta = 0 \tag{3.2.39}$$

where we solve for the two solutions r_a^+ and r_d^- by meshing the two quadratic solutions given by the equations. This give us two smooth paths for the positive critical point



$$r_d(\theta) = \frac{\cosh(\gamma) - \sqrt{\cosh^2(\gamma) - \frac{\cosh(\gamma)\theta}{\sin(\theta)}}}{\cos(\theta)}; \quad \theta \in [0, \pi) \tag{3.2.40}$$

$$r_a(\theta) = \frac{\cosh(\gamma) + \sqrt{\cosh^2(\gamma) - \frac{\cosh(\gamma)\theta}{\sin(\theta)}}}{\cos(\theta)}; \quad \theta \in [0, \pi) \tag{3.2.41}$$

With which we can construct both paths of constant imaginary part

$$\Gamma_d = \{\tilde{r}_d(\theta)\,\mathrm{e}^{\mathrm{i}\theta} : \theta \in (0, 2\pi)\}; \quad \Gamma_a = \{\tilde{r}_a(\theta)\,\mathrm{e}^{\mathrm{i}\theta} : \theta \in (0, 2\pi)\} \tag{3.2.42}$$

We need now to determine the Steepest Descent Path. In this case is more clear what the path should be the one passing through the critical point $\mathrm{e}^{-\gamma}$. Then, repeating the process we did on Case 1 - changing the path of integration - we can make the substitution

$$\begin{aligned}
\mathrm{He}_n(2\sqrt{N}\cosh(\gamma)) &= C_n^0 \int_\Gamma \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta && (3.2.43)\\
&= C_n^0 \int_{\Gamma_d} \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta && (3.2.44)\\
&\overset{(Steepest)}{=} C_n^0\,\mathrm{e}^{N(\phi_\alpha(\beta_c^-))}\left[\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}}\frac{\mathrm{e}^{\mathrm{i}(\theta(\beta_c^-))}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}} + \mathcal{O}(N^{-\frac{3}{2}})\right] && (3.2.45)\\
&= C_n^0\,\mathrm{e}^{N(\phi_\alpha(\beta_c^-))}\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}}\frac{\mathrm{e}^{\mathrm{i}(\theta(\beta_c^-))}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}}\left(1 + \mathcal{O}(N^{-1})\right) && (3.2.46)\\
&= \mathrm{e}^{N(\phi_\alpha(\beta_c^-))}\frac{(N-1)!N^{\frac{1}{2}(1-N)}}{2\pi\mathrm{i}}\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}}\frac{\mathrm{i}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}}\left(1 + \mathcal{O}(N^{-1})\right) && (3.2.47)\\
&= \mathrm{e}^{N(\phi_\alpha(\beta_c^-)-1)+1}\frac{(N-1)^{N-\frac{1}{2}}N^{-\frac{N}{2}}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}}\left(1 + \mathcal{O}(N^{-1})\right) && (3.2.48)\\
&= \mathrm{e}^{N(-1)+1}\frac{(N-1)^{N-\frac{1}{2}}N^{-\frac{N}{2}}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}}\left(1 + \mathcal{O}(N^{-1})\right) && (3.2.49)\\
&= \mathrm{e}^{-N\left(\frac{\mathrm{e}^{-2\gamma}}{2}+\gamma+1\right)+1}\frac{(N-1)^{N-\frac{1}{2}}N^{-\frac{N}{2}}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}}\left(1 + \mathcal{O}(N^{-1})\right) && (3.2.50)
\end{aligned}$$

Similarly, for the other case

$$\mathrm{He}_{n-1}(2\sqrt{N}\cosh(\gamma)) = \frac{\mathrm{e}^{-N\left(\frac{\mathrm{e}^{-2\gamma}}{2}+\gamma+1\right)}}{\mathrm{e}^{-\left(\frac{\mathrm{e}^{-2\gamma}}{2}+\gamma-2\right)}} \frac{(N-1)^{-\frac{N-1}{2}}}{\sqrt{|1-\mathrm{e}^{2\gamma}|}} \left(1+\mathcal{O}(N^{-1})\right) \tag{3.2.51}$$

Case 3

Let's start with a sketch of the proof for this behavior.

1. **Step One**: Write the integral for $N = n+1$

$$\mathrm{He}_n(\sqrt{N}\alpha) = C_n^0 \int_\Gamma \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta \tag{3.2.52}$$

   where $\phi_\alpha(\beta) = \beta^2 - 2\alpha\beta + \ln(\beta)$;

2. **Step two**: The function $\phi_\alpha(\beta)$ has a double critical point at 2. The steepest descent curve is defined by $Im(\phi_\alpha(z) - \phi_\alpha(2)) = 0$;

3. **Step three**: Deform the contour such that the only contribution for the integral comes from the critical point . Computing the Taylor approximation to the third order we have

$$\phi_\alpha(\beta) = \phi_\alpha(\sqrt{2}/2) + \phi_\alpha^{(3)}(\sqrt{2}/2)(\beta-\beta_c)^3 + \mathcal{O}(\beta^4) \tag{3.2.53}$$

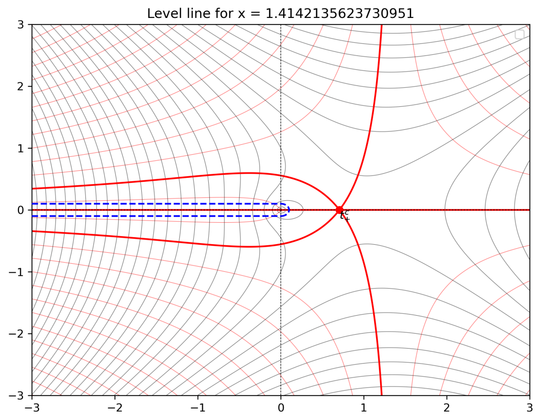
   and perform the change fo variables $\lambda = N^{\frac{1}{3}}(\beta-\beta_c)$ to obtain

$$\mathrm{He}_n(\sqrt{N}\alpha) \approx \tilde{C_n} \int_\Gamma \mathrm{e}^{\lambda^3}\,\mathrm{d}\lambda \tag{3.2.54}$$

4. **Step four** Recognize the Airy Function and write

$$\mathrm{He}_n(\sqrt{N}\alpha) \propto Ai(0) \tag{3.2.55}$$

Figure 3.4: Figure for case 3



We start with the expression determined in (3.3.4), working on the edge, we will set $x = 2\sqrt{N} + \xi N^{-\frac{1}{6}}$, that is, $\alpha = 2 + \xi N^{-\frac{2}{3}}$. Remember the expression

$$\mathrm{He}_n(\sqrt{N}\alpha) = C_n^0 \int_\Gamma \mathrm{e}^{-N\left(\frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta)\right)}\,\mathrm{d}\beta. \tag{3.2.56}$$

Applying the scale above, we have

$$\phi_{2+\xi N^{-\frac{2}{3}}}(\beta) = -(2+\xi N^{-\frac{2}{3}})\beta + \frac{1}{2}\beta^2 + \ln(\beta) = \phi_2(\beta) - \xi N^{-\frac{2}{3}}\beta, \tag{3.2.57}$$

such that substituting it into the integral

$$\mathrm{He}_n(\sqrt{N}(2+\xi N^{-\frac{2}{3}})) = C_n^0 \int_\Gamma \mathrm{e}^{-N\left(\phi_2(\beta) - \xi\beta N^{-\frac{2}{3}}\right)}\,\mathrm{d}\beta. \tag{3.2.58}$$

Now, it is clear from the previous cases that when we evaluate the integral, the contributions from the critical point $\beta_c = 1$ should be exponentially bigger - that is, the asymptotic of the integral can be deduces from the values around this point. We know that

$$\phi_2^{(0)}(1) = -\frac{3}{2} \tag{3.2.59}$$

$$\phi_2^{(1)}(1) = 0 \tag{3.2.60}$$

$$\phi_2^{(2)}(1) = 0 \tag{3.2.61}$$

$$\phi_2^{(3)}(1) = 2 \tag{3.2.62}$$

such that we can define for a neighborhood of the critical point the Taylor expansion of ϕ given by

$$\phi_2(\beta) = \phi_2(1) + \frac{\phi_2^{(3)}(1)}{3!}(\beta - \beta_c)^3 + \mathcal{O}((\beta - \beta_c)^4) \tag{3.2.63}$$

such that for $\beta = \beta_c + uN^{-\frac{1}{3}}$, for u coming from $\infty\,\mathrm{e}^{\mathrm{i}\frac{4\pi}{3}}$ to $\infty\,\mathrm{e}^{\mathrm{i}\frac{2\pi}{3}}$, we write

$$\mathrm{He}_n(2N^{\frac{1}{2}} + \xi N^{-\frac{1}{6}}) = C_n^0\,\mathrm{e}^{\frac{3N}{2}} \int_{\langle} \mathrm{e}^{-N\left(\frac{1}{3}(\beta-\beta_c)^3 - \xi\beta N^{-\frac{2}{3}} + \mathcal{O}((\beta-\beta_c)^4)\right)}\,\mathrm{d}\beta \tag{3.2.64}$$

$$= C_n^0\,\mathrm{e}^{\frac{3N}{2}} \int_{\langle} \mathrm{e}^{-N\left(\frac{1}{3}(uN^{-\frac{1}{3}})^3 - \xi(\beta_c + uN^{-\frac{1}{3}})N^{-\frac{2}{3}} + \mathcal{O}((uN^{-\frac{1}{3}})^4)\right)}\,\mathrm{d}(uN^{-\frac{1}{3}}) \tag{3.2.65}$$

$$= C_n^0 N^{-\frac{1}{3}}\,\mathrm{e}^{-2\xi\beta_c N^{\frac{1}{3}}}\,\mathrm{e}^{\frac{3N}{2}} \int_{\langle} \mathrm{e}^{\frac{1}{3}u^3 - \xi u + \mathcal{O}(u^4 N^{-\frac{4}{3}})}\,\mathrm{d}u \tag{3.2.66}$$

Then we will have, remembering the definition of the Airy function 3.2.2,

$$\mathrm{He}_n(2N^{\frac{1}{2}} + \xi N^{-\frac{1}{6}}) = \tilde{C}_N \int_{\langle} \mathrm{e}^{\left(\frac{u^3}{3} - \xi u\right)}\,\mathrm{d}u(1 + o(1)) = \tilde{C}_N Ai(\xi)(1 + o(1)), \tag{3.2.67}$$

for $\tilde{C}_N = (N-1)!N^{\frac{1}{6}-\frac{N}{2}}\,\mathrm{e}^{-2\xi\beta_c N^{\frac{1}{3}} + \frac{3N}{2}} = \sqrt{2\pi}(N-1)^{\frac{1}{2}} N^{1/6-N/2}\,\mathrm{e}^{\left(1/2 + N^{-1} - 2\xi\beta_c N^{-2/3}\right)}(1 + \mathcal{O}(N^{-1}))$. The result follow from the asymptotic for the Airy function given in 3.2.1. Again, we can change variables to express

$$\mathrm{He}_{n-1}(2\sqrt{N} + N^{-1/6}\xi) = \sqrt{2\pi}(N-2)^{\frac{1}{2}}(N-1)^{4/6-N/2}\,\mathrm{e}^{\left(1/2 + (N-1)^{-1} - 2\xi\beta_c(N-1)^{-2/3}\right)}\,Ai(\xi)(1 + o(1)) \tag{3.2.68}$$

3.3 Kernels

When dealing with the asymptotic of the j.p.d.f we will actually consider the Kernel. This is the object described in 1.3.3. However, we will actually work with the simplification given by 1.3.4. The first step is to consider the following expression

$$\mathrm{He}_n(\alpha\sqrt{N} + yN^{-1/2}), \tag{3.3.1}$$

which will allow us to take the limit to get $K_n^{(2)}(x, x)$.

3.3.1 Sine Kernel

The computations don't change too much. Consider the expression

$$\mathrm{He}_n(\sqrt{N}\alpha + N^{-1/2}y) = \frac{(N-1)!N^{\frac{1}{2}(1-N)}}{2\pi\mathrm{i}} \int_{\Gamma} \mathrm{e}^{-N(-\alpha\beta + \frac{1}{2}\beta^2 + \ln(\beta)) + y\beta}\,\mathrm{d}\beta \tag{3.3.2}$$

$$= C_n^0 \int_{\Gamma} \mathrm{e}^{-N\phi_\alpha(\beta) + y\beta}\,\mathrm{d}\beta \tag{3.3.3}$$

$$= C_n^0 \int_{\Gamma} \mathrm{e}^{-N\phi_\alpha(\beta)}\,g_y(\beta)\mathrm{d}\beta \tag{3.3.4}$$

with

$$\phi_\alpha(\beta) = \frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta); \quad g_y(\beta) = \mathrm{e}^{y\beta} \tag{3.3.5}$$

The analysis remains the same - the difference now being that we have a function $g_y(\beta)$ that enters in the steepest descent expression. Setting our case for $-2\sqrt{N} < x < 2\sqrt{N}$, or, equivalently, $-2 < \alpha < 2$, we remember that

$\beta_c^{\pm} = \mathrm{e}^{\pm \mathrm{i}\gamma}$. Now, with this, and setting $\Delta = 2\sqrt{N}\cos(\gamma) + N^{-1/2}y$,

$$\begin{aligned}
\mathrm{He}_n(\Delta) &= C_n^0 \int_\Gamma \mathrm{e}^{-N\phi_\alpha(\beta)}\, g_y(\beta)\mathrm{d}\beta = C_n^0 \left[\int_{\Gamma_d^-} \mathrm{e}^{-N\phi_\alpha(\beta)}\, g_y(\beta)\mathrm{d}\beta + \int_{\Gamma_d^+} \mathrm{e}^{-N\phi_\alpha(\beta)}\, g_y(\beta)\mathrm{d}\beta\right] \\
&\overset{(Steepest)}{=} C_n^0 \left(\mathrm{e}^{N(\phi_\alpha(\beta_c^+))} \left[\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{g_y(\beta_c^+)\,\mathrm{e}^{\mathrm{i}(\theta(\beta_c^+))}}{\sqrt{2\sin(\gamma)}} + \mathcal{O}(N^{-\frac{3}{2}})\right] + \mathrm{e}^{N(\overline{\phi_\alpha(\beta_c^+)})} \left[\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{g_y(\beta_c^-)\,\mathrm{e}^{\mathrm{i}(\overline{\theta(\beta_c^+)})}}{\sqrt{2\sin(\gamma)}} + \mathcal{O}(N^{-\frac{3}{2}})\right]\right) \\
&= C_n^0 \sqrt{\frac{2\pi}{2N\sin(\gamma)}} \left(\mathrm{e}^{N(\phi_\alpha(\beta_c^+))} \left[g_y(\beta_c^+)\,\mathrm{e}^{\mathrm{i}(\theta(\beta_c^+))} + \mathcal{O}(N^{-1})\right] + \mathrm{e}^{N(\overline{\phi_\alpha(\beta_c^+)})} \left[g_y(\beta_c^-)\,\mathrm{e}^{\mathrm{i}(\overline{\theta(\beta_c^+)})} + \mathcal{O}(N^{-1})\right]\right) \\
&= C_n^0 \sqrt{\frac{\pi}{N\sin(\gamma)}}\, \mathrm{e}^{y\cos(\gamma)} \left(\mathrm{e}^{N(\phi_\alpha(\beta_c^+)) + \mathrm{i}(\theta(\beta_c^+) + y\sin(\gamma))} + \mathrm{e}^{N(\overline{\phi_\alpha(\beta_c^+)}) + \mathrm{i}(\overline{\theta(\beta_c^+)} - y\sin(\gamma))}\right)(1 + \mathcal{O}(N^{-1})) \\
&= C_n^0 \sqrt{\frac{\pi}{N\sin(\gamma)}}\, \mathrm{e}^{y\cos(\gamma)}\, \mathrm{e}^{N(-3\cos^2(\gamma) - \frac{1}{2})}\, 2\mathrm{i}\sin\left(\theta(\beta_c^+) + y\sin(\gamma) - N\left(\frac{\sin(2\gamma)}{2} + \gamma\right)\right)(1 + \mathcal{O}(N^{-1})) \\
&= \frac{(N-1)!N^{\frac{1}{2}(1-N)}}{2\pi\mathrm{i}} \sqrt{\frac{\pi}{N\sin(\gamma)}}\, \mathrm{e}^{y\cos(\gamma)}\, \mathrm{e}^{N(-3\cos^2(\gamma) - \frac{1}{2})}\, 2\mathrm{i}\sin\left(\theta(\beta_c^+) + y\sin(\gamma) - N\left(\frac{\sin(2\gamma)}{2} + \gamma\right)\right)(1 + \mathcal{O}(N^{-1})) \\
&= \sqrt{\frac{2}{\sin(\gamma)}} N^{-\frac{N}{2}} (N-1)^{\left(N - \frac{1}{2}\right)}\, \mathrm{e}^{N\left(-3\cos^2(\gamma) + \frac{1}{2}\right) + 1 + y\cos(\gamma)} \sin\left(\theta(\beta_c^+) + y\sin(\gamma) - N\left(\frac{\sin(2\gamma)}{2} + \gamma\right)\right)(1 + \mathcal{O}(N^{-1}))
\end{aligned} \tag{3.3.6}$$

and for $N-1$,

$$\mathrm{He}_{n-1}(\Delta) = \sqrt{\frac{2}{\sin(\gamma)}} \frac{(N-2)^{N-\frac{3}{2}}}{(N-1)^{\frac{N-1}{2}}} \frac{\mathrm{e}^{N\left(-3\cos^2(\gamma) + \frac{1}{2}\right) + 1 + y\cos(\gamma)}}{\mathrm{e}^{\left(-3\cos^2(\gamma) + \frac{1}{2}\right)}} \sin\left(\theta(\beta_c^+) + y\sin(\gamma) - (N-1)\left(\frac{\sin(2\gamma)}{2} - \gamma\right)\right)(1 + \mathcal{O}(N^{-1})) \tag{3.3.7}$$

Remembering the expression for the Kernel, taking $\Delta_x = \sqrt{N}\alpha + N^{-1/2}x$ and $\Delta_y = \sqrt{N}\alpha + N^{-1/2}y$

$$K_n^{(2)}(\Delta_x, \Delta_y) = \frac{\sqrt{w(\Delta_x)w(\Delta_y)}}{r_{n-1}^{(2)}} \frac{H_n(\Delta_x)H_{n-1}(\Delta_y) - H_{n-1}(\Delta_x)H_n(\Delta_y)}{\Delta_x - \Delta_y} \tag{3.3.8}$$

$$= W_{(N,x,y)} \frac{H_n(\Delta_x)H_{n-1}(\Delta_y) - H_{n-1}(\Delta_x)H_n(\Delta_y)}{(\sqrt{N}\alpha + N^{-1/2}x) - (\sqrt{N}\alpha + N^{-1/2}y)} \tag{3.3.9}$$

$$= W_{(N,x,y)} N^{1/2} \frac{H_n(\Delta_x)H_{n-1}(\Delta_y) - H_{n-1}(\Delta_x)H_n(\Delta_y)}{x - y} \tag{3.3.10}$$

with

$$W_{(N,x,y)} = \frac{\sqrt{w(\sqrt{N}\alpha + N^{-1/2}x)w(\sqrt{N}\alpha + N^{-1/2}y)}}{r_{n-1}^{(2)}} \tag{3.3.11}$$

$$= \left(\frac{\exp\left(-\frac{1}{2}\left(\sqrt{N}\alpha + N^{-1/2}x\right)^2\right)}{\sqrt{2\pi}} \frac{\exp\left(-\frac{1}{2}\left(\sqrt{N}\alpha + N^{-1/2}y\right)^2\right)}{\sqrt{2\pi}}\right)^{-\frac{1}{2}} \tag{3.3.12}$$

$$= \frac{1}{\sqrt{2\pi}} \left(\exp\left\{-\frac{1}{2}\left(4N\cos(\gamma) + 4x\cos(\gamma) + x^2N^{-1} + 4N\cos(\gamma) + 4y\cos(\gamma) + y^2N^{-1}\right)\right\}\right)^{\frac{1}{2}} \tag{3.3.13}$$

$$= \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-2N\cos(\gamma)} \exp\left\{-\left(2\cos(\gamma)(x+y) + \frac{1}{2N}(x^2 + y^2)\right)\right\} \tag{3.3.14}$$

Note that

$$H_n(\Delta_x)H_{n-1}(\Delta_y) \propto \sin\left(\theta(\beta_c^+) + x\sin(\gamma) - N\left(\frac{\sin(2\gamma)}{2} + \gamma\right)\right) \sin\left(\theta(\beta_c^+) + y\sin(\gamma) - (N-1)\left(\frac{\sin(2\gamma)}{2} - \gamma\right)\right) \tag{3.3.15}$$

and

$$H_{n-1}(\Delta_x)H_n(\Delta_y) \propto \sin\left(\theta(\beta_c^+) + y\sin(\gamma) - N\left(\frac{\sin(2\gamma)}{2} + \gamma\right)\right)\sin\left(\theta(\beta_c^+) + x\sin(\gamma) - (N-1)\left(\frac{\sin(2\gamma)}{2} - \gamma\right)\right) \tag{3.3.16}$$

therefore,

$$H_n(\Delta_x)H_{n-1}(\Delta_y) - H_{n-1}(\Delta_x)H_n(\Delta_y) \propto \sin\left(\frac{\sin(2\gamma)}{2} - \gamma\right)\sin(\sin(\gamma)(x-y)) \tag{3.3.17}$$

Then, letting $x = x\sin(\gamma)$ and $y = y\sin(\gamma)$

$$K_n^{(2)}(\Delta_x, \Delta_y) = W_{(N,x,y)}N^{1/2}C\,\mathrm{e}^{\frac{1}{2}\sin(2\gamma)(x+y)}\sin\left(\frac{\sin(2\gamma)}{2} - \gamma\right)\frac{\sin(x-y)}{x-y} \tag{3.3.18}$$

where

$$C = \left(\sqrt{\frac{2}{\sin(\gamma)}}N^{-\frac{N}{2}}(N-1)^{\left(N-\frac{1}{2}\right)}\mathrm{e}^{N\left(-3\cos^2(\gamma)+\frac{1}{2}\right)+1}\right)\left(\sqrt{\frac{2}{\sin(\gamma)}}\frac{(N-2)^{N-\frac{3}{2}}}{(N-1)^{\frac{N-1}{2}}}\frac{\mathrm{e}^{N\left(-3\cos^2(\gamma)+\frac{1}{2}\right)+1}}{\mathrm{e}^{\left(3\cos^2(\gamma)-\frac{1}{2}\right)}}\right) \tag{3.3.19}$$

$$= \frac{2}{\sin(\gamma)}\left(\frac{(N-1)(N-2)^{2-3N^{-1}}}{N}\right)^{\frac{N}{2}}\frac{\mathrm{e}^{2N\left(-3\cos^2(\gamma)+\frac{1}{2}\right)+1}}{\mathrm{e}^{-3\cos^2(\gamma)+\frac{1}{2}}} \tag{3.3.20}$$

and

$$W_{(N,x,y)} = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-2N\cos(\gamma)}\exp\left\{-\left(\sin(2\gamma)(x+y) + \frac{\sin^2(\gamma)}{2N}(x^2+y^2)\right)\right\} \tag{3.3.21}$$

Here, we get to the point of interest.

Definition 3.3.1. *The Kernel*

$$\mathcal{S}(x,y) := \frac{\sin(x-y)}{x-y} \tag{3.3.22}$$

is called the Sine Kernel.

With this definition, we can now say that

Theorem 3.3.2. *For the conditions $-2\sqrt{N} < x < 2\sqrt{N}$, or, equivalently, $-2 < \alpha < 2$, we remember that $\beta_c^{\pm} = \mathrm{e}^{\pm i\gamma}$. Set $\Delta_x = 2\sqrt{N}\cos(\gamma) + \frac{\sin(\gamma)}{\sqrt{N}}x$ and $\Delta_y = 2\sqrt{N}\cos(\gamma) + \frac{\sin(\gamma)}{\sqrt{N}}y$. Then, we have that*

$$K_n^{(2)}(\Delta_x, \Delta_y) = \frac{2N^{\frac{1}{2}}}{\sin(\delta)\sqrt{2\pi}}\exp\left\{-\frac{1}{2}\left(\sin(2\gamma)(x+y) + \frac{\sin^2(\gamma)}{N}(x^2+y^2)\right) - 3\left(\cos(\gamma) - \frac{1}{2}\right)\right\}$$
$$\left(\frac{(N-1)(N-2)^{2-\frac{3}{N}}}{N}\right)^{\frac{N}{2}}\mathcal{S}(x,y)(1+\mathcal{O}(N^{-1})) \tag{3.3.23}$$

3.3.2 Airy Kernel

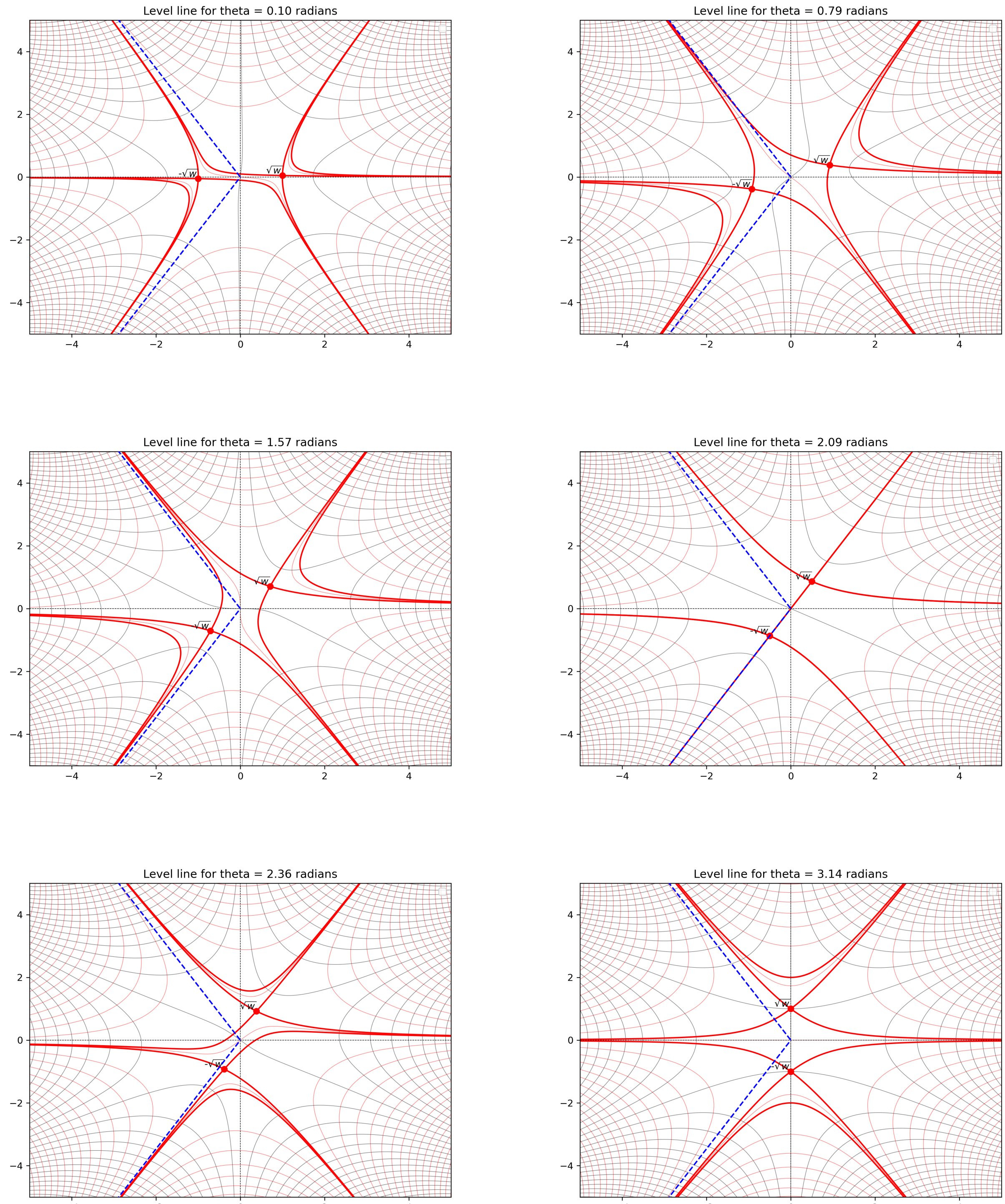
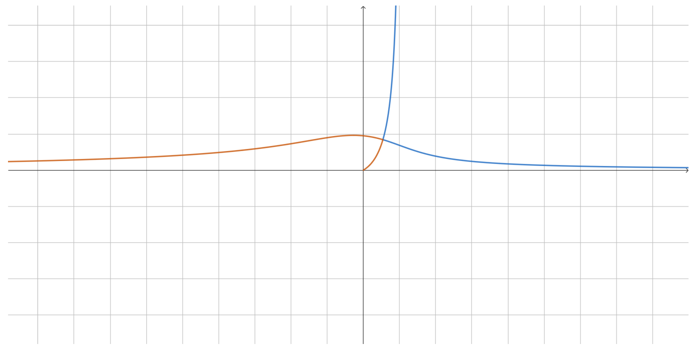
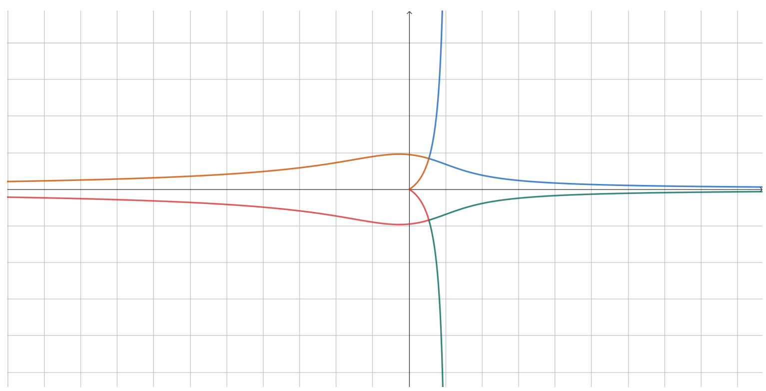
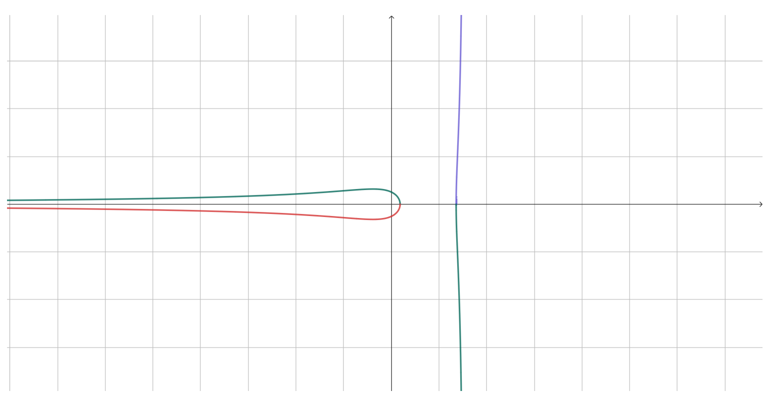


Figure 3.1: Scenarios for θ







Chapter 4

Miscellaneous

4.1 Laplace Transform

4.2 Fourrier Transform

We start by defining what we call a Fourrier Series. Take $f(x)$ to be a a-periodic function, i.e., $f(x+a) = f(x),\ \forall\ x \in \mathbb{R}$. How to characterize such a function? We may try to specify the values it takes on the interval but that's uncountable big. We can also try to approximate it somehow, one such approach is to use geometric periodic functions. We write:

$$f(x) = \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{2\pi n x}{a}\right) + \sum_{m=1}^{\infty} \beta_m \cos\left(\frac{2\pi m x}{a}\right) = \sum_{-\infty}^{\infty} f_n \, \mathrm{e}^{2\pi \mathrm{i} n \frac{x}{a}}$$

Note that such a summation must be a-periodic as all functions used are a-periodic. Of course, not all functions are guarantee to be expressed in such a way. A know class that works, and usually enough for our purposes, are square-integrable functions, for which

$$\int_{-a/2}^{a/2} |f(x)|^2 \mathrm{d}x$$

exists and is finite. The concept we have is useful but limited if used as such. We want to extend such a concept for functions that are not a-periodic - but how? Define a family of functions $\{f_a(x) | a \in \mathbb{R}^+\}$ such that $f_a(x)$ is a-periodic and $\lim_{a\to\infty} f_a(x) = f(x)$. We have that every function f_a in the family can be written as a Fourrier Series as it is periodic, so we write:

$$f_a(x) = \sum_{-\infty}^{\infty} f_{an} \, \mathrm{e}^{\mathrm{i} k_n x} : \ k_n = n\Delta k; \Delta k = \frac{2\pi}{a}.$$

And as we know about the limit, we can write

$$\begin{aligned} f(x) &= \lim_{a\to\infty} \left[\sum_{-\infty}^{\infty} f_{an} \, \mathrm{e}^{\mathrm{i} k_n x} \right] = \lim_{a\to\infty} \left[\sum_{-\infty}^{\infty} \frac{\Delta k}{2\pi} \, \mathrm{e}^{\mathrm{i} k_n x} \left(\frac{2\pi f_{an}}{\Delta k} \right) \right] \\ &= \int_{-\infty}^{\infty} \frac{\mathrm{d}k}{2\pi} \, \mathrm{e}^{\mathrm{i}kx} \, F(k); \quad \text{where} \quad F(k) = \lim_{a\to\infty} \left[\frac{2\pi f_{an}}{\Delta k} \right]. \end{aligned} \tag{4.2.1}$$

where we call $F(x)$ the Fourrier Tranform of the function $f(x)$. We can also define the inverse transform

$$\begin{aligned} F(k) &= \lim_{a\to\infty} \left[\frac{2\pi}{\Delta k} f_{an}(x) \right] \\ &= \lim_{a\to\infty} \left[\frac{2\pi}{\frac{2\pi}{a}} \frac{1}{a} \int_{-a/2}^{a/2} \mathrm{d}x \, \mathrm{e}^{-\mathrm{i} k_n x} \right] \\ &= \int_{-\infty}^{\infty} \mathrm{d}x \, \mathrm{e}^{-\mathrm{i}kx} \, f(x) \end{aligned} \tag{4.2.2}$$

With this definition we can make explicit some properties of the transform, namely:

1. Linearity: Given $f(x)$ and $g(x)$, with its respective $F(x)$ and $G(x)$ Fourrier Tranforms we have that

$$af(x) + bg(x) \rightarrow_{FT} aF(k) + bG(k);$$

2. Translation: Given $f(x)$ translated by b we write

$$f(x+b) \rightarrow_{FT} \mathrm{e}^{ikb}\, F(k);$$

3. Derivative: Given $f(x)$ we have that the derivative is written

$$\frac{\mathrm{d}}{\mathrm{d}x} f(x) \rightarrow_{FT} \mathrm{i}kF(k)$$

4.2.1 On Differential Equations

Take the damped harmonic oscillator subjected to an additional force $f(t)$. The equation of motion is given by

$$\frac{\mathrm{d}^2 x(t)}{\mathrm{d}t} + 2\gamma \frac{\mathrm{d}x(t)}{\mathrm{d}t} + \omega_0^2 x(t) = \frac{f(t)}{m}.$$

To solve for $x(t)$ we first take the Fourrier Tranform

$$-\omega^2 X(\omega) - 2\mathrm{i}\gamma\omega X(\omega) + \omega_0^2 X(\omega) = \frac{F(\omega)}{m}$$

where now we can reorganize to solve for $X(\omega)$.

$$X(t) = \frac{\frac{F(\omega)}{m}}{-\omega^2 - 2\mathrm{i}\gamma\omega + \omega_0^2}.$$

Now it is a simple case of taking the inverse of the Fourrier Transform

$$x(t) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \frac{\frac{\mathrm{e}^{-i\omega t}\, F(\omega)}{m}}{-\omega^2 - 2\mathrm{i}\gamma\omega + \omega_0^2}; \quad \text{where } F(\omega) = \int_{-\infty}^{\infty} \mathrm{d}t\, \mathrm{e}^{\mathrm{i}\omega t}\, f(t)$$

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